

Computational investigation of temperature dependent spin relaxation of point defects

- A theoretical study of the T_1 time

*Beräkningsundersökning av temperaturberoende spinrelaxation
hos punktdefekter*

Oskar Bollner

Supervisor : Oscar Bulancea Lindvall
Examiner : Joel Davidsson

© Oskar Bollner, 2026

© ⓘ This work is licensed under Creative Commons Attribution 4.0 International. To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Abstract

As quantum technology advances towards practical applications, researchers are actively broadening the search for scalable material platforms. While there are many physical realizations of qubits, this thesis focuses on point defects in semiconductors, defect qubits, as they have proven long qubit lifetimes on the order of milliseconds at room temperature. All qubit systems have a characteristic relaxation time T_1 , which describes how long an excited state survives before decaying toward thermal equilibrium and consequently bounds the achievable phase coherence time T_2 . Therefore, the T_1 time is a vital component for properly understanding defect systems. The T_1 time in point defects is strongly dependent on the temperature because of the increased phonon occupation. In this thesis, we model the relaxation rate of two point defects; the well-studied nitrogen-vacancy (NV) center in diamond and the novel chlorine-vacancy (ClV) center in 4H-SiC, which displays telecom-band emission frequencies making it appropriate for fiber-optic communication applications. However, accurate modeling of defect spin dynamics is complicated by spin contamination in unrestricted DFT, an artifact addressed in this work. The T_1 relaxation time was derived from first- and second-order spin-phonon coupling via finite differences. We apply two methods for calculating the ZFS, one including all valence-band and defect spin orbitals, and one restricted to the unpaired defect electrons. The results show that there is a strong connection between spin contamination and symmetry-breaking phonon modes transforming as the E irreducible representation (irrep) of the C_{3v} point group, suggesting that spin contamination artificially enhances the contribution of E -symmetric modes to the relaxation rate. It was found that at high temperatures the T_1 time is strongly dominated by two-phonon Raman processes, which supports previous research. The ClV center exhibits stronger spin-phonon coupling, especially in the low-energy region, giving rise to additional allowed transitions not present in the NV center, due to the lower phonon energies of 4H-SiC. Therefore, the ClV center has a shorter room-temperature T_1 time $58.7\mu\text{s} \leq T_1 \leq 2.1\text{ms}$ compared to the NV center $7.5\text{ms} \leq T_1 \leq 45.5\text{ms}$. These results identify spin contamination as a significant source of error in defect spin-phonon modeling, and provide a framework for benchmarking T_1 in wide-bandgap qubit candidates.

Acknowledgments

I would like to thank my supervisor Oscar Bulancea Lindvall for helping me unravel the mathematical soup of group theory and complex physics of spin-phonon interactions. I would also like to thank my examiner Joel Davidsson for the opportunity of writing my thesis here at IFM. An extra thanks goes to Petter Johansson for asking somewhat annoying but much needed questions making me reconsider and restructure some of the methods i have used in the thesis.

Contents

Abstract	iii
Acknowledgments	iv
Contents	v
List of Figures	vii
List of Tables	ix
List of Terms	x
List of Abbreviations	xi
1 Introduction	1
1.1 Color centers	2
1.2 Spin property calculations	2
1.3 Aim	2
2 Theory	4
2.1 Point defects	4
2.2 Phonon modes	5
2.3 Spin Hamiltonian	10
2.4 Interaction Hamiltonian	12
2.5 Transition rates in a phonon bath	14
2.6 Density functional theory	17
2.7 T_1 Relaxation time	19
3 Method	21
3.1 Computational details	21
3.2 Calculating the relaxation rate	23
3.3 Calculation of the T_1 relaxation time	24
4 Results	25
4.1 NV center in diamond	25
4.2 CIV center in 4H-SiC	33
5 Discussion	39
5.1 Predicted stability of NV center in diamond	39
5.2 Predicted stability of CIV center in 4H-SiC	39
5.3 Spin contamination and the connection to symmetry	40
5.4 Analysis of the IPR	40
5.5 Method	40
5.6 The work in a wider context	41

5.7 Outlook	41
6 Conclusion	42
Bibliography	43
A Mathematical derivations	49
A.1 Zero Field splitting	49
A.2 Time-dependent perturbation theory	50

List of Figures

2.1	Defect types. (a) Vacancy. (b) Substitution. (c) Interstitial.	4
2.2	Energy diagram of the NV center in diamond and CIV center in 4H-SiC. The band gap is not to scale.	5
2.3	Illustration of a n mass system of springs. Each atom can either move in phase, out of phase, or be stationary, giving n oscillatory modes.	6
2.4	Phonon mode after applying symmetry operations. (a) The original phonon mode, or application of the identity operator which has an overlap of 1. (b) Phonon mode after applying a rotation of 120° around the principal axis, yielding an overlap of 1. (c) Phonon mode after mirroring the mode in the σ_v plane, with an overlap of -1	8
2.5	Example of a local A_1 mode, seen along the principal axis of C_{3v} . $u_{\kappa,l}$ stands for the mass-weighted displacement of atom κ for phonon mode l	8
2.6	Illustration of a (a) localized mode with high IPR and (b) delocalized mode with low IPR.	10
2.7	Zero field splitting between states $ 0\rangle$ and $ \pm\rangle$ in the NV center in diamond [6].	11
2.8	Possible second order transitions between states $ m'_s\rangle$ and $ m_s\rangle$ in the case of $E_{m'_s} > E_{m_s}$. Figure inspired by Ref. [36].	17
3.1	Flowchart for the process of calculating the relaxation rate, from input of crystal structure data to relaxation rate calculation from ZFS coefficients.	21
4.1	The coupling coefficients and spectral function for first and second order phonons for 512 atom supercell. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	26
4.2	The coupling coefficients and spectral function for first and second order phonons for 64 atom supercell. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	27
4.3	The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 512 atom supercell NV center. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	28
4.4	The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 64 atom supercell NV center. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	29
4.5	Transition rates for a 512 atom supercell size NV center. (a) Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and (b) with spin corrected ZFS. (c) Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and (d) with spin corrected ZFS.	30

4.6	Transition rates for a 64 atom supercell size NV center. (a) Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and (b) with spin corrected ZFS. (c) Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and (d) with spin corrected ZFS.	31
4.7	The T_1 relaxation time, for the $m_s = 0$ polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.	32
4.8	The T_1 relaxation time, for the $m_s = +1$ polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.	32
4.9	The T_1 relaxation time, for the $m_s = -1$, polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.	33
4.10	The coupling coefficients and spectral function for first and second order phonons for 128 atom supercell CIV center in 4H-SiC. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	34
4.11	The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 128 supercell CIV center. (a) Results for first-order coupling with spin contamination and (b) with spin-corrected ZFS. (c) Second-order spin-phonon coupling with spin contamination and (d) with spin-corrected ZFS.	35
4.12	Transition rates for a 128 atom supercell size CIV center. (a) Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and (b) with spin corrected ZFS. (c) Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and (d) with spin corrected ZFS.	36
4.13	The T_1 relaxation time, for the $m_s = 0$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.	37
4.14	The T_1 relaxation time, for the $m_s = +1$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.	37
4.15	The T_1 relaxation time, for the $m_s = -1$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.	38

List of Tables

2.1	Character table for C_{3v}	8
2.2	Multiplication table for irreps of the C_{3v} point group.	9
4.1	T_1 relaxation time at room temperature (300 K) of a 64 and 512 atom NV center in diamond.	33
4.2	T_1 relaxation time at room temperature (300 K) of a 128 atom kk CIV center in 4H-SiC.	38
4.3	T_1 relaxation time at 0 K of a 128 atom kk CIV center in 4H-SiC.	38

LIST OF TERMS

color center A point defect in a crystal (such as a vacancy, impurity, or complex) that introduces electronic states within the band gap, giving rise to optical absorption and often visible coloration. Nitrogen-vacancy centres in diamond are a prominent example. 2

identity element An operator or element in a group that, when combined with any other element under the group operation, leaves that element unchanged. 7

inverse element For a given element in a group, the unique element that when combined with it yields the identity element. 7

phonon Energy quanta of an elastic wave in a crystal lattice. 2, 4

qubit A quantum bit is the fundamental unit of quantum information. Unlike a classical bit that can only be 0 or 1, a qubit can exist in a superposition of the two basis states $|0\rangle$ and $|1\rangle$, enabling quantum parallelism and entanglement. Physical implementations include superconducting qubits, trapped ions, and spin states of point defects such as the nitrogen-vacancy center in diamond. 1, 39, 42

spin contamination The unwanted mixing of higher spin states into an unrestricted wavefunction (e.g., unrestricted Hartree-Fock or unrestricted DFT), causing the expectation value of the total spin operator $\langle \hat{S}^2 \rangle$ to deviate from the ideal value $S(S + 1)$ and degrading calculated properties such as energies and spin densities. 18

LIST OF ABBREVIATIONS

- CIV** A point defect in 4H-SiC similar to the NV center where a chlorine atom replaces a carbon atom next to a Si vacancy. 2, 5, 7, 12, 15, 21, 22, 25, 33–36, 39–42
- DFT** A quantum mechanical method for calculating the electronic structure of many-body systems, using functionals of the electron density. 1, 2, 4, 17, 18, 21, 22, 40–42
- HSE** A screened hybrid functional that mixes a fraction of exact Hartree–Fock exchange with PBE exchange, improving band gap predictions. 18
- IPR** A measure of the degree of localisation of a quantum state. Larger values indicate stronger localisation. 9, 25, 27, 33, 34, 40, 42
- irrep** A representation of a group that cannot be decomposed into a direct sum of smaller representations. 7, 9, 12, 42
- ISC** The process of transitioning between spin mutliplets, caused by spin-orbit coupling and phonon interactions. 5
- MLIP** An interatomic potential constructed by fitting machine-learning models to first-principles energies and forces, enabling large-scale atomistic simulations with near-DFT accuracy. 41
- NV** A point defect in diamond consisting of a substitutional nitrogen atom adjacent to a lattice vacancy. It acts as an optically active spin qubit. 2, 4, 5, 7, 12, 15, 19, 21, 22, 25, 29, 31, 33, 35, 39–42
- ODE** ordinary differential equation 4, 19
- PAW** projector augmented wave 22
- PBE** A widely used exchange-correlation functional of the generalized gradient approximation (GGA) in DFT. 18, 22
- pymatgen** An open-source Python library for materials analysis, providing tools for electronic structure, crystal structure, and phase diagrams. 22
- VASP** A widely used computer program for atomic scale materials modelling based on DFT, using pseudopotentials and a plane-wave basis set. 21, 22
- ZFS** The lifting of degeneracy of spin states in the absence of an external magnetic field. 2, 3, 5, 10–13, 15, 19, 21–23, 25, 33, 35, 39, 40, 42, 49

INTRODUCTION

The ages of humanity are closely related to the materials used. In history, we can see how technologically advanced a civilisation was by looking at the materials in their tools and how they used them, such as the Stone age, Bronze age, and Iron age. One could say that we are currently living in the information age which is built upon the technology of semiconductors, like silicon. Semiconductors are of the most common materials used in many industries today, from microprocessors in electronics to synthetic silicone polymers in construction and biology. We are today approaching the next technological age, the quantum age, and semiconductors might get another use in the field of quantum technology.

Quantum technology is a growing field because of its potential in secure communication [1], high precision sensors [2, 3], and computation [4]. Current research is still at the stage of finding and optimizing the quantum material platform. Finding new materials has been and is still a cumbersome process requiring a lot of testing with expensive lab equipment. However, nowadays, one can use additional approaches to accelerate the process of screening materials and iterating designs via materials simulations at the quantum mechanical level using methods such as density functional theory (DFT) on high performance computing clusters - increasing the pace at which new materials can be discovered by many orders of magnitude. It is still, however, a computationally heavy task for classical computers to simulate the rules of quantum mechanics, especially larger structures such as molecules and solids. This process could be made exponentially faster if implemented on a quantum computer [4], but to build one, we first need to create stable and scalable quantum bits (*qubits*).

A qubit is, like a classical bit, a binary unit of information which can be represented by the numbers 0 and 1. The difference between a qubit and classical bit is that a qubit has the qualities of superposition and entanglement, the former meaning that the probability of both bit values (0 and 1) can be non-zero at the same time, and multiple qubits can become entangled, meaning their joint state has correlations that cannot be explained classically. For an entangled state, the qubit holds information both about itself and the other entangled qubit. In a fully entangled state, the two entangled qubits are indistinguishable from each other. There are many ways to create a qubit as it only requires a two-level quantum system, e.g., polarization of light using horizontal polarization as 0 and vertical as 1 [5] or the spin state of an electron as spin up (1) or spin down (0) [6]. The difficulty lies in making a qubit resistant to noise, as the information stored in quantum systems is particularly sensitive to external interactions and easily dissipates.

An important metric of a qubit is the rate at which information, or state coherence, stored in a single qubit is lost. This time interval is often referred to as the coherence time T_2 . This arises from random noise causing loss of phase information of the system. Since most qubits exchange energy with its environment in some way, it will also inevitably relax to thermal equilibrium after some time T_1 referred to as the relaxation time [7]. The T_1 time is therefore an upper bound to the stability of the qubit and sets a theoretical limit of $T_2 \leq 2T_1$ [8–10]. Another important metric of a quantum platform is the *gate* time, which is the time it takes for one qubit operation. Some of the techniques to create a qubit include quantum dots, which exhibit $T_1 \sim 1$ s [11], $T_2 \sim 10 \mu\text{s}$ to $100 \mu\text{s}$ and gate times of a couple hundred nanoseconds. Superconducting qubits have a $T_2 \approx T_1 \approx 100 \mu\text{s}$ and gate times of ~ 100 ns, they are however difficult to scale because of manufacturing variations impacting the compatibility between qubits. Superconducting qubits also require sub-kelvin temperatures making them difficult to use outside of lab environments. There are also trapped ion qubits with T_1 times of a

couple of hours up to a few days, $T_2 \geq 50$ s and single qubit gate times of 10 ns to 5 μ s. The high relaxation and coherence times are because of the isolated design of trapped ions, simultaneously, this isolation prohibits qubit manipulation and readout [12]. Color centers, which are the subject of this thesis, have $T_1 \sim 1$ ms, T_2 between several hundred nanoseconds to microseconds and gate times of ~ 50 ns at room temperature [7].

1.1 Color centers

A material defect in the crystal lattice modifies the electronic structure of the material, this can cause the defect to emit light of a specific wavelength. Defects of this kind are therefore named *color centers*. They can introduce new electron orbitals in the bandgap of the material leading to a spin configuration acting as an isolated two-state system, which can be used as a qubit. The dangling-bond electrons in the defect are then used to perform quantum computations by optically induced spin flips, often in the microwave range [7, 13]. An advantage of color centers is that they have been shown to have long coherence times of up to several milliseconds at room temperature [13, 14], in contrast to many other quantum platforms which require very low temperatures for high coherence times [7]. A fundamental limitation of color center qubits is crystal vibrational modes driving spin relaxation as the vibrational activity increases with temperature. Therefore, modelling the temperature dependant T_1 time is important for screening new materials suitable for quantum applications. It is, however, a difficult computational task to model these quantum mechanical systems theoretically using methods such as DFT.

The negatively charged nitrogen-vacancy (NV)[−] center in diamond [15] is highly interesting because of its beneficial properties for quantum computation and sensing [16, 17]. It consists of a diamond crystal modified by substituting a carbon atom with a nitrogen and removing the nearest neighbor carbon atom, and adding an extra electron. Multiple previous studies have been conducted to analyse the interaction between the defect electron spins and phonons of the surrounding crystal lattice which impact the spin-phonon relaxation rate [18]. This system is well studied in regards to the spin-phonon coupling and qubit performance investigated in this work [19, 20]. In contrast, a recently proposed system is the chlorine-vacancy (ClV) center in 4H-SiC, which consist of a substitutional chlorine on carbon and Si vacancy. It is believed to be the source for various novel observed emissions in the telecom band range, making it easier to integrate with modern communication standards [21, 22]. Since it can be made from SiC wafers, which already have well established production infrastructure, producing ClV centers is not as big a task as creating new infrastructure for other defects such as the NV center.

1.2 Spin property calculations

To calculate spin properties of point defects, one uses the electron wavefunctions calculated from DFT simulations. However, orbitals obtained in unrestricted spin-DFT can generally not be described as single-particle eigenfunctions of the original Hamiltonian. Most relevant is the fact that the mixing of the orbitals as basis states of the S^2 operator causes erroneous evaluation of spin-based properties such as the zero field splitting (ZFS) [23]. This phenomenon is termed *spin contamination* [24]. Elimination of such errors are an active field of investigation, especially in finding efficient procedures compatible with high-throughput approaches for defect discovery [25, 26]. More on this in Chapter 2.

1.3 Aim

This thesis aims to calculate the spin-phonon T_1 time for the NV center in diamond and ClV center in 4H-SiC and investigate how to improve current computational methods. More

specifically analyse how the localization of phonon modes affects T_1 and if simulation time can be shortened by removing spin contamination from the ZFS calculations without substantially affecting predicted T_1 values.

The theoretical concepts one needs to understand in this thesis are explained in this chapter. The theory begins by describing point defects and their relevant features to quantum applications. Concepts such as phonons, group theory and symmetry are explained to lay a foundation for later sections of the theory explaining the spin-phonon interaction Hamiltonian and how the crystal symmetry can be utilized to simplify calculations. Fermi's golden rule, based on time dependent perturbation theory, is applied to the spin-phonon coupling Hamiltonian to extract transition rates between the spin sublevels of the defect. These rates are then used to solve a system of ordinary differential equations (ODEs) to find the T_1 relaxation time. The chapter also provides a short description of DFT and explains the emergence of spin contamination.

2.1 Point defects

Crystals are seldom perfect, they contain defects that disrupt the periodic lattice. Defects are commonly classified by dimensionality, i.e. point defects (0D), line defects (1D), planar defects (2D), and volume defects (3D). Point defects, the focus of this thesis, can form naturally during crystal growth or be introduced artificially by irradiation or ion implantation. They exist in all crystalline materials and their concentration is often at the ppb or ppm scale. Three fundamental types of point defects can be distinguished: vacancies, substitutions, and interstitials.

A *vacancy* is a removal of an atom in the crystal lattice, leaving an empty space. A *substitution* is a replacement of an atom with another element in the same lattice point. An *interstitial* is an insertion of an atom in-between the lattice points, see Figure 2.1.

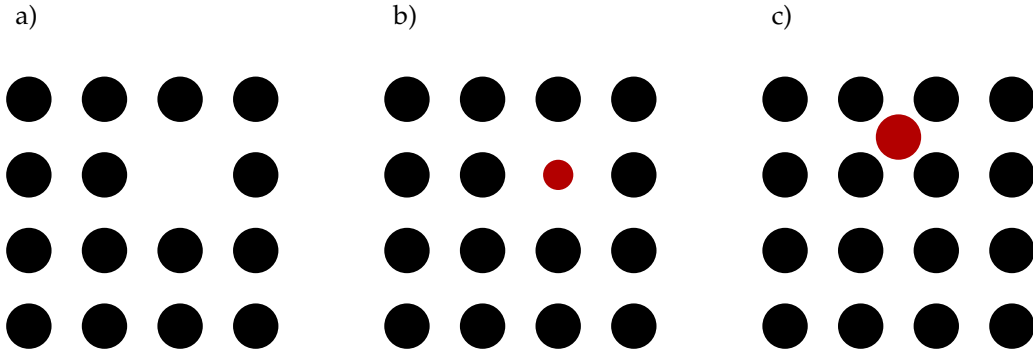


Figure 2.1: Defect types. (a) Vacancy. (b) Substitution. (c) Interstitial.

Point defects are of extra interest in wide bandgap materials such as diamond, silicon carbide, etc. because they can introduce deep, well isolated, electron orbitals in the bandgap. Electrons occupying these deep orbitals have a low probability of spontaneously relaxing or thermally exciting, and remain stable during manipulation, which is useful for quantum technologies. One point defect with such properties is the previously mentioned negatively charged NV⁻ center in diamond. It introduces 6 electrons, four of which occupy orbitals inside the band gap. Two electrons occupy the lowest-lying orbitals, denoted by a_1 , and the remaining two occupy the degenerate e_x and e_y orbitals, see Figure 2.2. These two e -electrons are *unpaired*, meaning each occupies an orbital without a partner of opposite spin, which

gives the defect a net magnetic moment and leads to a spin triplet ($S = 1$) ground state and a low-lying spin singlet ($S = 0$) excited state. Because of spin-spin dipolar interactions, the spin triplet has an energy splitting between the $m_s = 0$ and $m_s = \pm 1$ states of 2.87 GHz [6]. This energy splitting will be explained further in Section 2.3. Additionally, the spin system can be used as a qubit as it can be initialized, controlled and read out optically. To use the spin states, one has to initialize or *polarize* the spin state into one of the spin sublevels. Preferably to the spin sublevel with the highest coherence and relaxation times.

Initialization of the defect qubit involves optical spin polarization. The process for the NV center is as follows: optical excitation from a_1 to the e orbitals conserves spin. If the initial state was $m_s = 0$, the electron can decay directly back by emitting a photon. If the initial state was $m_s = \pm 1$, the excited state has a high probability of crossing to a metastable singlet state via intersystem crossing (ISC). From this singlet, the system preferentially relaxes into the $m_s = 0$ sublevel of the ground-state triplet. Repetition of this cycle thus optically polarizes the NV⁻ center into $m_s = 0$ [13, 27], preparing the qubit.

The CIV center can be constructed in four different variations, the hh , hk , kh and kk variants. These different variants have different symmetries as the SiC structure is layered. The hh and kk variants are constructed along the symmetry axis, and the hk and kh variants are off-axis defects reducing the crystal symmetry [21]. The variant studied in this thesis is the kk CIV defect. The kk defect configuration has been noted to have a ZFS of 1.04 GHz to 1.34 GHz [28]. The optical spin polarization of CIV is similar to the NV center as it involves ISC between the singlet and triplet states, with the difference that it is predicted to favour the $m_s = \pm 1$ state [29].

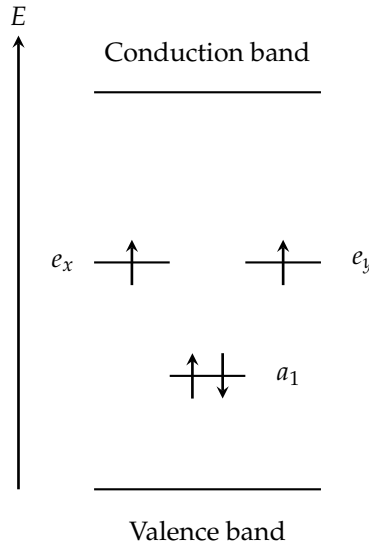


Figure 2.2: Energy diagram of the NV center in diamond and CIV center in 4H-SiC. The band gap is not to scale.

2.2 Phonon modes

Solid materials are not static, they have vibrational modes which can get thermally excited. These modes are called phonons and are the quantization of energy for elastic waves. In a finite three dimensional crystal structure with n atoms, there are $3n$ modes of oscillation since each atom has three degrees of freedom [30]. A simple example illustrating possible modes is a one dimensional chain of n atoms like in Figure 2.3. In one dimension, each atom has one degree of freedom, so the chain has n oscillatory modes.

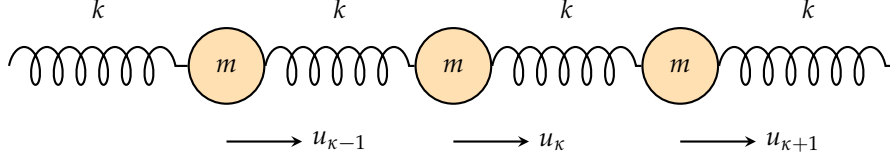


Figure 2.3: Illustration of a n mass system of springs. Each atom can either move in phase, out of phase, or be stationary, giving n oscillatory modes.

Phonons can be seen as states described by harmonic oscillators of different energies

$$\hat{H}_{ph} = \sum_l^{3N} \hbar \omega_l (\hat{b}_l^\dagger \hat{b}_l + \frac{1}{2}), \quad (2.1)$$

where $\hat{b}_l$ and $\hat{b}_l^\dagger$ is the phonon annihilation and creation operators respectively and ω_l is the frequency of phonon mode l .

The occupation of these states follow Bose-Einstein statistics and the average total number of occupied states at a given temperature $n(T)$ is defined as

$$\langle n(T) \rangle = \sum_l^{3N} \frac{1}{e^{\hbar \omega_l / k_B T} - 1}. \quad (2.2)$$

Since this thesis concerns point groups of defect centers, we restrict this analysis to the Γ point ($\mathbf{q} = 0$) which conserves the full C_{3v} symmetry. To describe the atomic positions in a three dimensional periodic crystal structure, we introduce the vector $\mathbf{r}_\kappa^a$ which is the position of atom κ in unit cell a . The phononic displacement $\mathbf{u}_\kappa^a(t)$ is introduced as a deviation from the equilibrium position $\mathbf{R}_\kappa^a(t) = \mathbf{r}_\kappa^a + \mathbf{u}_\kappa^a(t)$. The total energy of the crystal with respect to ionic displacement about the equilibrium is described as

$$E(\mathbf{r}_\kappa^a) = E_0 + \frac{\partial E}{\partial \mathbf{u}_\kappa^a} \mathbf{u}_\kappa^a + \frac{\partial^2 E}{\partial \mathbf{u}_\kappa^a \partial \mathbf{u}_{\kappa'}^{a'}} \mathbf{u}_\kappa^a \mathbf{u}_{\kappa'}^{a'} + \dots \quad (2.3)$$

The first-order term is zero as the forces acting on the crystal vanish at equilibrium. In the harmonic approximation, terms of order greater than two are neglected. From Eq. (2.3), we define the second-order force constants $\Phi_{\alpha,\beta}^{\kappa,\kappa'}(a)$ where α and β are the Cartesian components. From this expansion, we define the dynamical matrix $\mathcal{D}_{\alpha\beta}^{\kappa\kappa'}$ as

$$\mathcal{D}_{\alpha\beta}^{\kappa\kappa'} = \frac{1}{\sqrt{m_\kappa m_{\kappa'}}} \sum_a^N \Phi_{\alpha,\beta}^{\kappa,\kappa'}(a) \quad (2.4)$$

where N is the number of unit cells. The equation of motion for the phonons is written as

$$\sum_{\kappa',\beta} \mathcal{D}_{\alpha\beta}^{\kappa\kappa'} e_{\kappa',\beta,l} = \omega_l^2 e_{\kappa,\alpha,l} \quad (2.5)$$

where $e_{\kappa,\alpha,l}$ and l are the Cartesian components of the eigenvectors of the dynamical matrix and the phonon band index, respectively. These eigenvectors are later used to characterize the phonon symmetry. The Cartesian atomic displacement is then defined as

$$\mathbf{u}_\kappa^a(t) = \left(\frac{\hbar}{2Nm_\kappa} \right)^{1/2} \sum_l \omega_l^{-1/2} \left[\hat{b}_l e^{-i\omega_l t} + \hat{b}_l^\dagger e^{i\omega_l t} \right] \mathbf{e}_{\kappa,l}. \quad (2.6)$$

A total dimensionless crystal perturbation Q_l of phonon mode l can then be defined from the atomic displacement vector in Eq. (2.6) as

$$Q_l = A \sqrt{\frac{2\omega_l}{\hbar}} \quad (2.7)$$

where A is the amplitude of the mass-weighted displacement in units $\text{\AA}\sqrt{\text{a.m.u.}}$.

Symmetry and group theory

In higher dimensions and more complex structures, the phonons can have interesting symmetry properties which can be used to simplify computation. In order to describe these symmetry properties mathematically, one can utilize group theory.

Definition 2.2.1: Group

A group is a set G , together with a binary operation $\circ$, satisfying the following group axioms:

Closure: $\forall a, b \in G, \exists c \in G : a \circ b = c$

Associativity: $\forall a, b, c \in G$ one has $a \circ (b \circ c) = (a \circ b) \circ c$.

Identity element: $\exists e \in G$ s.t $\forall a \in G$, one has $a \circ e = a$ and $e \circ a = a$. Such an element is unique and is called the *identity element* of G .

Inverse element: $\forall a \in G, \exists b \in G$ s.t $a \circ b = e$ and $b \circ a = e$, where e is the identity element of G . For each a there exists a unique element b called the *inverse element* of a and is often denoted a^{-1} .

In the context of crystal structures, a space group can be defined as the set of all symmetry operations mapping the structure on to itself, i.e rotations, reflections, translations, etc. A subgroup of the space group is the point group, which holds one point of the crystal lattice fixed under all operations in the group, i.e., translations. In order to apply these operations on a crystal structure they can be represented as matrices such that

$$D(AB) = D(A)D(B) \quad (2.8)$$

where A and B are elements in the group and D is the matrix representation of it. $D(A)$ is called a *representation* of the group. If a representation can be brought to a block-diagonal form, it is reducible, meaning there exists some operation P which applied to the representation yields

$$D'(a) = P^{-1}D(a)P = \begin{pmatrix} D_1(a) & & & \\ & D_2(a) & & \\ & & \ddots & \\ & & & D_N(a) \end{pmatrix} \quad (2.9)$$

where the blocks $D_i(a)$, which can not be decomposed further, are the irreducible representations (irreps) of the group. Because the phonon Hamiltonian is invariant under all operations of the crystals space group, its solutions transform according to the representations of that group. Consequently, each phonon mode can be classified by the irreps according to which its atomic displacements transform.

The structure of the NV center and kk variant of the CIV center have C_{3v} symmetry. This means that the phonons can be described by the irreps of the C_{3v} point group, A_1 , A_2 and E . The C_{3v} point group has three operations, E , C_3 , and σ_v , which is the identity, a 120° rotation around the symmetry axis (z-axis), and reflection in the xz-plane respectively, see Figure 2.4. The modes transforming like the E irreps exist in pairs of E_x and E_y modes which due to the restriction of group theory are degenerate. An example of an A_1 mode, which is invariant under all group operations, can be seen in Figure 2.5

The character χ_l of a phonon mode l is identified by evaluating

$$\chi_l = \text{Tr}(\mathbf{U}_l \mathbf{P} \mathbf{U}_l^T) \quad (2.10)$$

where $\mathbf{U}_l$ is a $N \times 3$ tensor describing the displacement direction of each atom in a specific phonon mode l and $\mathbf{P}$ is a point group operation, depicted in Figure 2.4.

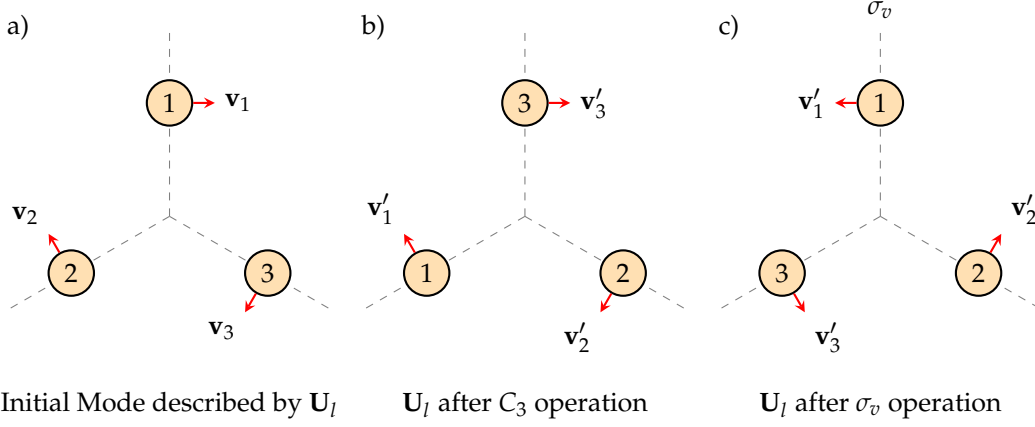


Figure 2.4: Phonon mode after applying symmetry operations. **(a)** The original phonon mode, or application of the identity operator which has an overlap of 1. **(b)** Phonon mode after applying a rotation of 120° around the principal axis, yielding an overlap of 1. **(c)** Phonon mode after mirroring the mode in the σ_v plane, with an overlap of -1 .

The mode symmetry is identified by matching the character of each mode to the characters in the C_{3v} character table in Table 2.1.

Table 2.1: Character table for C_{3v}

C_{3v}	E	$2C_3$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E	2	-1	0

E modes are a special case as they come in pairs and the sum of their characters, not their individual contributions, for σ_v needs to be zero.

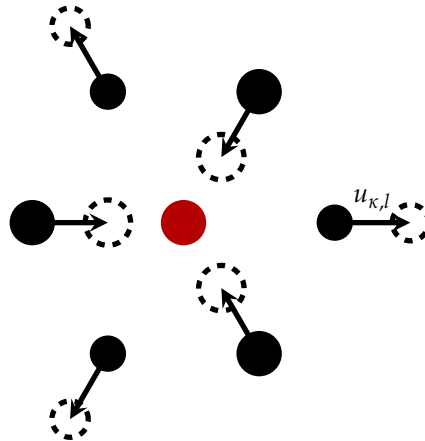


Figure 2.5: Example of a local A_1 mode, seen along the principal axis of C_{3v} . $u_{\kappa,l}$ stands for the mass-weighted displacement of atom κ for phonon mode l .

Cartesian products of irreducible representations

When there are multiple phonon modes in the system, their resulting atomic displacement can be described as a product of irreps, following Table 2.2.

Table 2.2: Multiplication table for irreps of the C_{3v} point group.

	A_1	A_2	E	
A_1	A_1	A_2	E	A_1
A_2	A_2	A_1	E	A_2
E	E	E	$A_1 \oplus A_2 \oplus E$	E
	A_1	A_2	E	

For modes of character A_1 , E_x , and E_y , the dimensionless phonon coordinate operators are denoted $\hat{R}_l$, $\hat{X}_l$, and $\hat{Y}_l$ and are defined by $\hat{R}_l = (\hat{b}_{l_R} + \hat{b}_{l_R}^\dagger)/\sqrt{2}$, $\hat{X} = (\hat{b}_{l_X} + \hat{b}_{l_X}^\dagger)/\sqrt{2}$, and $\hat{Y} = (\hat{b}_{l_Y} + \hat{b}_{l_Y}^\dagger)/\sqrt{2}$. In second-order phonon combinations, the combined phonon normal coordinates must obey the same symmetry rules as the displacements; therefore we can construct the second-order normal coordinates as

$$\begin{cases} A_1 : \hat{R}^2, \\ E_x : \hat{X}^2 - \hat{Y}^2, \\ E_y : \hat{X}\hat{Y} + \hat{Y}\hat{X}. \end{cases} \quad (2.11)$$

Inverse participation ratio

Phonon mode displacement can be distributed differently in the crystal around the defect. In order to quantify the locality of displacement, one can use the inverse participation ratio (IPR). This ratio will tell how gathered a field is around its center and is defined as

$$\text{IPR}_l = \frac{\sum_{\kappa} |\mathbf{e}_{i\kappa,l}|^4}{(\sum_{\kappa} |\mathbf{e}_{\kappa,l}|^2)^2} \quad (2.12)$$

where n is the number of atoms in the supercell and $|\mathbf{e}_{\kappa,l}|^2$ is the amplitude squared of the eigenvector of the dynamical matrix for atom κ and mode l . The localization is represented in Figure 2.6.

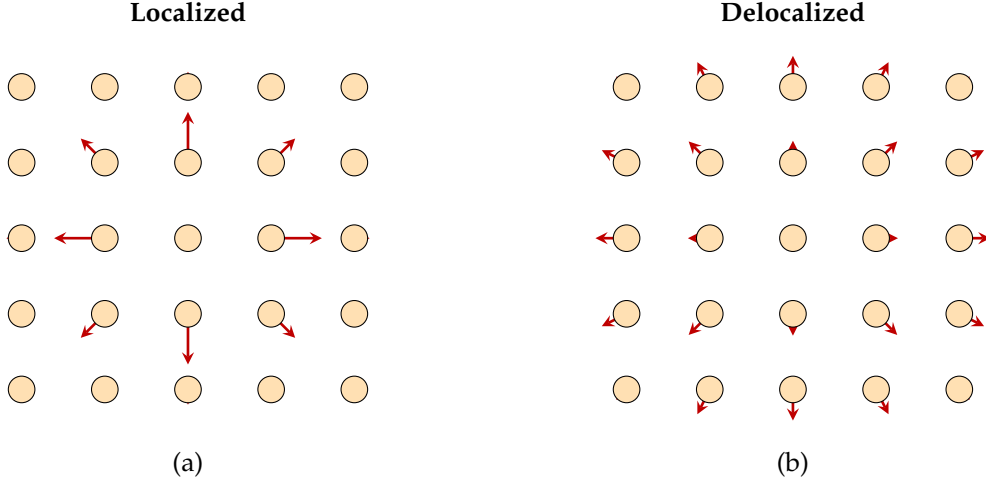


Figure 2.6: Illustration of a **(a)** localized mode with high IPR and **(b)** delocalized mode with low IPR.

2.3 Spin Hamiltonian

A spin Hamiltonian describes the dynamics of effective electronic spin, nuclear spin, and an external magnetic field. The eigenvalues of the spin Hamiltonian determine the allowed energy levels of the system. The total spin Hamiltonian for an effective defect electron spin in a bath of nuclear spins of the host material, in the presence of an external magnetic field, can be described by the form [31]

$$\begin{aligned} \hat{H}_s = & \hat{\mathbf{S}} \cdot \mathbf{D} \cdot \hat{\mathbf{S}} + \mu_B \hat{\mathbf{S}} \cdot \mathbf{g} \cdot \vec{\mathbf{B}} + \sum_{\nu} \hat{\mathbf{I}}^{\nu} \cdot \gamma^{\nu} \cdot \vec{\mathbf{B}} \\ & + \sum_{\nu} \hat{\mathbf{S}} \cdot \mathbf{A}^{\nu} \cdot \hat{\mathbf{I}}^{\nu} + \sum_{\nu, \mu} \hat{\mathbf{I}}^{\mu} \cdot \mathbf{Q}^{\mu\nu} \cdot \hat{\mathbf{I}}^{\nu} + \sum_{\nu, \mu} \hat{\mathbf{S}}^{\mu} \cdot \mathbf{J}^{\mu\nu} \cdot \hat{\mathbf{S}}^{\nu}. \end{aligned} \quad (2.13)$$

The first term is the intra-defect electronic spin-spin interaction, also referred to as the ZFS term, where $\hat{\mathbf{S}}$ is the effective electronic spin vector operator and $\mathbf{D}$ the ZFS-tensor. The spin Hamiltonian has many terms but the only term of interest in this thesis is the ZFS.

The remaining terms consist of $\mathbf{g}$ the g-tensor, $\vec{\mathbf{B}}$ the magnetic field vector, $\hat{\mathbf{I}}$ the nuclear spin vector operator, γ^{ν} the gyromagnetic ratio of nuclei ν , $\mathbf{A}^{\nu}$ the hyperfine tensor, $\mathbf{Q}^{\mu\nu}$ the nuclear dipole-dipole coupling tensor, or in the case of $\mu = \nu$ it reduces to the nuclear quadrupole interaction, and $\mathbf{J}^{\mu\nu}$ is the inter-defect electronic spin-spin coupling tensor, i.e. $\mu \neq \nu$. The sums are taken over the interactions between different particles and defects. The terms describe the different spin interactions of the system. The second and third terms are the electronic and nuclear Zeeman interactions. The fourth is the electronic spin interactions with nuclear spins also known as hyperfine interactions, fifth, intra-defect nuclear spin-spin interaction, and lastly the electronic spin-spin interaction between adjacent defects.

The Zeeman terms are neglected since only the spin-phonon coupling is investigated, not the effects of an external field. The intra-defect nuclear spin-spin interaction can be neglected as the ratio between the nucleon magnetic moment and the electron magnetic moment is $\mu_N/\mu_e < 10^{-3}$. The electronic spin-spin interaction between electrons in adjacent defects can also be neglected since they are relatively small compared to the ZFS due to the distance between neighbouring defects being $\gg 10 \text{ \AA}$ and the dipole interaction decays as $1/r^3$. The hyperfine interaction is also neglected due to it being small compared to the ZFS [32, 33] and would also introduce more complexity to the model which require more advanced computational methods not studied in this thesis. The expression for the spin

Hamiltonian is then simply

$$\hat{H}_s = \hat{H}_{ZFS}. \quad (2.14)$$

Zero field splitting

If the defect has more than one unpaired electron, every unpaired electron will interact via dipole-dipole interactions. The ZFS Hamiltonian can be described by a total spin $\hat{\mathbf{S}} = \sum_i^N \hat{\mathbf{S}}_i$, under the conditions that the spin-orbit coupling is relatively weak and separable from the spin-spin interaction [34]. The ZFS term of the spin hamiltonian can then be written as

$$\hat{H}_{ZFS} = \hat{\mathbf{S}} \cdot \mathbf{D} \cdot \hat{\mathbf{S}} = \sum_{a,b} \hat{S}_a \hat{S}_b D_{ab} \quad (2.15)$$

where $\mathbf{D}$ is a 3×3 tensor which describes the total spin interaction of the defect and the subscripts a, b refer to the Cartesian coordinate axes. The components of the ZFS tensor can be calculated by integrating over all possible configurations of the interacting electrons described by the two-particle density matrix

$$\rho_2(\mathbf{r}_1, \mathbf{r}_2) = N(N-1) \int \dots \int |\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{x}_3, \dots, \mathbf{x}_N)|^2 d\mathbf{x}_3 \dots d\mathbf{x}_N \quad (2.16)$$

$$D_{ab} = \frac{1}{2} \frac{\mu_0 g_e^2 \mu_B^2}{4\pi} \int \rho_2(r_1, r_2) \frac{r^2 \delta_{ab} - 3r_a r_b}{r^5} dr_1 dr_2 \quad (2.17)$$

where μ_0 , g_e , and μ_B are the permeability in vacuum, free electron g-factor $g_e \approx 2$, and Bohr magneton respectively [34]. Here r_a and r_b refer to the Cartesian components of $\mathbf{r} = |\mathbf{r}_1 - \mathbf{r}_2|$. From the form of the dipole-dipole interaction in Eq. (2.17), it follows that $\mathbf{D}$ must be symmetric and traceless. Since the ZFS tensor is symmetric we can find a basis diagonalizing $\mathbf{D}$ such that

$$\mathbf{D} = \begin{pmatrix} D_{xx} & 0 & 0 \\ 0 & D_{yy} & 0 \\ 0 & 0 & D_{zz} \end{pmatrix}. \quad (2.18)$$

Given that we are working in the diagonal basis for $\mathbf{D}$ and $\mathbf{D}$ is traceless we can simplify the hamiltonian to (see Section A.1 for derivation)

$$\hat{H}_{ZFS} = D \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) + E \left(\hat{S}_x^2 - \hat{S}_y^2 \right) \quad (2.19)$$

where $D = \frac{3}{2} D_{zz}$ and $E = \frac{1}{2} (D_{xx} - D_{yy})$. For axially symmetric crystals, the hamiltonian is simply

$$\hat{H}_{ZFS} = D \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right). \quad (2.20)$$

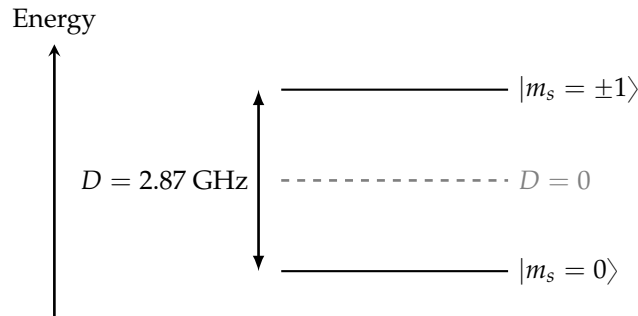


Figure 2.7: Zero field splitting between states $|0\rangle$ and $|\pm\rangle$ in the NV center in diamond [6].

The spin Hamiltonian is affected by the spin density of the crystal structure which is impacted by ionic positions. Therefore, the spin Hamiltonian will depend on phonon excitations in the crystal.

2.4 Interaction Hamiltonian

The total Hamiltonian describing a spin-phonon system can be written in three contributing terms

$$\hat{H} = \hat{H}_{ph} + \hat{H}_s(\mathbf{Q} = 0) + \hat{H}_{s-ph} \quad (2.21)$$

where $\hat{H}_{ph}$ is the phonon harmonic oscillator Hamiltonian, $\hat{H}_s(\mathbf{Q} = 0)$ is the spin Hamiltonian evaluated without phonon perturbations, and $\hat{H}_{s-ph}$ is the spin-phonon interaction term. An expression for the spin-phonon interaction is not known explicitly and therefore one typically performs a Taylor expansion of the spin Hamiltonian $\hat{H}_s$ with respect to a lattice perturbation Q_l along the phonon normal coordinates and neglect the zeroth-order term [18, 35]

$$\hat{H}_{s-ph} = \sum_l \frac{\partial \hat{H}_s}{\partial Q_l} Q_l + \sum_{ll'} \frac{\partial^2 \hat{H}_s}{\partial Q_l \partial Q_{l'}} Q_l Q_{l'} + \dots \quad (2.22)$$

where the derivatives are referred to as the spin-phonon coupling coefficients.

Spin-phonon coupling coefficients

To calculate the spin-phonon coupling coefficients, we look at the change of the spin Hamiltonian with respect to lattice perturbations in an ON-basis using the symmetry axis of the defect as the z-axis. This basis diagonalizes the D-tensor for the relaxed NV and CIV center. If we apply this rotation to the D-tensor of the perturbed structure, one can utilize the properties of C_{3v}

$$\tilde{\mathbf{D}} = \mathbf{R}^T \mathbf{D} \mathbf{R}. \quad (2.23)$$

We then express the D-tensor as a Taylor series with respect to the total dimensionless phonon perturbation Q_l from phonon mode l

$$\tilde{\mathbf{D}}(\mathbf{Q}) = \tilde{\mathbf{D}}(\mathbf{Q} = 0) + \sum_l \left. \frac{d\tilde{\mathbf{D}}}{dQ_l} \right|_{\mathbf{Q}=0} Q_l + \frac{1}{2} \sum_{ll'} \left. \frac{d^2 \tilde{\mathbf{D}}}{dQ_l dQ_{l'}} \right|_{\mathbf{Q}=0} Q_l Q_{l'} + \dots \quad (2.24)$$

The derivatives are evaluated using finite differences, explained further in Chapter 3.

Recall, from Section 2.2, that the displacement of phonon modes transform according to the irreps of the C_{3v} point group. This property is used to simplify computation of $|0\rangle \leftrightarrow |\pm 1\rangle$ single- and $|+1\rangle \leftrightarrow |-1\rangle$ double-quantum transitions of the spin-phonon coupling coefficients. Since the Hamiltonian must be invariant to symmetry operations of the point group, the Hamiltonians of the NV and CIV centers transform as the irrep A_1 of the C_{3v} point group. Because of this symmetry property we can expand Eq. (2.24) in terms where the dimensionless phonon perturbations obey the symmetry of A_1 and $E_{\{X,Y\}}$. The dimensionless coordinates are then paired with symmetry adapted spin operators in the ZFS to construct fully A_1 symmetric terms. Similar to Eq. (2.11), we can construct combinations of spin operators which transform according to the irreps of C_{3v} as

$$\begin{cases} A_1 : \hat{S}_z^2 - \frac{S(S+1)}{3}, \\ E_x : \hat{S}_x^2 - \hat{S}_y^2, \\ E_y : \hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x. \end{cases} \quad (2.25)$$

Following this reasoning, one can expand the ZFS tensor as

$$\begin{aligned}
 \tilde{\mathbf{D}}(\mathbf{R}) = & \begin{pmatrix} -\frac{1}{3}D & & \\ & -\frac{1}{3}D & \\ & & +\frac{2}{3}D \end{pmatrix} + \begin{pmatrix} -\frac{1}{3} & & \\ & -\frac{1}{3} & \\ & & +\frac{2}{3} \end{pmatrix} \sum_i 3V_{00}^i R_i \\
 & + \sum_l V_{+-}^l \left[\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} X_l + \begin{pmatrix} & 1 & \\ & & 1 \\ & & 0 \end{pmatrix} Y_l \right] \\
 & + \sum_l \sqrt{2}V_{0\pm}^l \left[\begin{pmatrix} & & 1 \\ & 0 & \\ 1 & & \end{pmatrix} X_l + \begin{pmatrix} 0 & & \\ & 1 & \\ & & 1 \end{pmatrix} Y_l \right] + \begin{pmatrix} -\frac{1}{3} & & \\ & -\frac{1}{3} & \\ & & +\frac{2}{3} \end{pmatrix} \sum_{ii'} 3V_{00}^{ii'} R_i R_{i'} \\
 & + \sum_l V_{+-}^{ll'} \left[\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} (X_l^2 - Y_l^2) + \begin{pmatrix} & 1 & \\ & & 1 \\ & & 0 \end{pmatrix} 2X_l Y_l \right] \\
 & + \sum_l \sqrt{2}V_{0\pm}^{ll'} \left[\begin{pmatrix} & & 1 \\ & 0 & \\ 1 & & \end{pmatrix} (X_l^2 - Y_l^2) + \begin{pmatrix} 0 & & \\ & 1 & \\ & & 1 \end{pmatrix} 2X_l Y_l \right] + \dots
 \end{aligned} \tag{2.26}$$

where R_i , X_l and Y_l are the dimensionless coordinates for the phonon modes, V_{00} , $V_{0\pm}$, and V_{+-} are the coupling coefficients of zero-, single-, and double-quantum transitions respectively. This expansion gives the Hamiltonian

$$\begin{aligned}
 \hat{\mathbf{S}} \cdot \tilde{\mathbf{D}} \cdot \hat{\mathbf{S}} = & D \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) \\
 \text{1st-order} \left\{ \begin{aligned} & + \sum_i 3V_{00}^i \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) \hat{R}_i \\ & + \sum_l V_{+-}^l \left[(\hat{S}_x^2 - \hat{S}_y^2) \hat{X}_l + (\hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x) \hat{Y}_l \right] \\ & + \sum_l \sqrt{2}V_{0\pm}^l \left[(\hat{S}_x \hat{S}_z + \hat{S}_z \hat{S}_x) \hat{X}_l + (\hat{S}_y \hat{S}_z + \hat{S}_z \hat{S}_y) \hat{Y}_l \right] \end{aligned} \right. \\
 \text{2nd-order} \left\{ \begin{aligned} & + \sum_i 3V_{00}^{ii} \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) \hat{R}_i^2 \\ & + \sum_l V_{+-}^{ll} \left[(\hat{S}_x^2 - \hat{S}_y^2) (\hat{X}_l^2 - \hat{Y}_l^2) + (\hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x) (\hat{X}_l \hat{Y}_l + \hat{Y}_l \hat{X}_l) \right] \\ & + \sum_l \sqrt{2}V_{0\pm}^{ll} \left[(\hat{S}_x \hat{S}_z + \hat{S}_z \hat{S}_x) (\hat{X}_l^2 - \hat{Y}_l^2) + (\hat{S}_y \hat{S}_z + \hat{S}_z \hat{S}_y) (\hat{X}_l \hat{Y}_l + \hat{Y}_l \hat{X}_l) \right] + \dots \end{aligned} \right.
 \end{aligned} \tag{2.27}$$

The spin-phonon coupling Hamiltonian can be expressed more compactly as

$$H_{s-ph} = \sum_i \left(\sum_{l \in \Gamma_i} V_i^l \hat{x}_l + \sum_{l \otimes l' \in \Gamma_i} V_i^{ll'} \hat{x}_l \hat{x}_{l'} \right) \hat{F}_i(\hat{\mathbf{S}}) \tag{2.28}$$

where $i = x, y, x', y', z$ is the spin label, $\Gamma_{x,y,x',y'} = E$ and $\Gamma_z = A_1$ are the irreducible representations of the C_{3v} point group. V_i^l and $V_i^{ll'}$ are the spin-phonon coupling coefficients and $\hat{x}_l = (\hat{b}_l + \hat{b}_l^\dagger) / \sqrt{2}$ are the dimensionless phonon coordinates expressed in second quantization. The spin operators $\hat{F}_i(\hat{\mathbf{S}})$ are the different symmetry adapted spin operators from Eq. (2.27), $\hat{F}_x(\hat{\mathbf{S}}) = \hat{S}_x^2 - \hat{S}_y^2$, $\hat{F}_y(\hat{\mathbf{S}}) = \hat{S}_x \hat{S}_y + \hat{S}_y \hat{S}_x$, $\hat{F}_{x'}(\hat{\mathbf{S}}) = \sqrt{2}(\hat{S}_x \hat{S}_z + \hat{S}_z \hat{S}_x)$, $\hat{F}_{y'}(\hat{\mathbf{S}}) = \sqrt{2}(\hat{S}_z \hat{S}_y + \hat{S}_y \hat{S}_z)$, and $\hat{F}_z(\hat{\mathbf{S}}) = 3 \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right)$. Or, expressed as matrices

$$\begin{aligned}
 \hat{F}_x(\hat{\mathbf{S}}) &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \hat{F}_y(\hat{\mathbf{S}}) &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \\
 \hat{F}_{x'}(\hat{\mathbf{S}}) &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}, & \hat{F}_{y'}(\hat{\mathbf{S}}) &= \begin{pmatrix} 0 & i & 0 \\ -i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \\
 \hat{F}_z(\hat{\mathbf{S}}) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix},
 \end{aligned} \tag{2.29}$$

with the basis vectors $|+1\rangle = (1 \ 0 \ 0)^T$, $|0\rangle = (0 \ 1 \ 0)^T$, and $|-1\rangle = (0 \ 0 \ 1)^T$. In this general form, we can see that we have no contributions from A_2^{ph} modes in the linear terms since the product $A_2^{ph} \otimes E^s = E$ and $A_2^{ph} \otimes A_1^s = A_2$, which is incompatible with the requirement of the Hamiltonian having A_1 character. However, in second order phonon interactions, we have the allowed contributions

$$A_2^{ph} \otimes A_2^{ph} \otimes A_1^s = A_1$$

and

$$A_2^{ph} \otimes E^{ph} \otimes E^s = A_1 \oplus A_2 \oplus E.$$

The superscript indicates phonon or spin symmetry.

2.5 Transition rates in a phonon bath

The defect state interacts with the lattice through a phonon bath, or reservoir. Using Fermi's golden rule taken to first-order in perturbation theory (see Section A.2), we can calculate the transition rates between two states

$$\Gamma_{f \leftarrow i} = \frac{2\pi}{\hbar} |\langle f | H | i \rangle|^2 \delta(E_f - E_i \pm (\hbar\omega)) \tag{2.30}$$

where $|i\rangle$ and $|f\rangle$ are the initial and final total states of the system (both spin and phonon state), and the Dirac delta is for energy conservation, the positive/negative sign in the delta function refers to absorption/emission of a phonon. To compare to previous research [19, 20, 36], the transition rates considered in this thesis are from the first-order phonon interaction taken to first- and second-order in perturbation theory and two-phonon interaction taken to first-order in perturbation theory. The sum of these contributions is referred to as the total transition rate between two states $|m_s\rangle$ and $|m'_s\rangle$ and is defined as

$$\Gamma_{m'_s \leftarrow m_s} = \sum_{i,j} \Gamma_{j, m'_s \leftarrow m_s}^{(i)} \tag{2.31}$$

where the subscript denotes the order in perturbation theory and the superscript denotes the order of spin-phonon interaction.

In first-order phonon interactions and first-order perturbation theory, there are two possible transitions. Either a phonon is absorbed and the spin transitions to a higher energy state, or, a phonon is emitted and the spin transitions to a lower energy state. If we denote $\Gamma_{1, m'_s \leftarrow m_s}^{(1), em/abs}$ as the transition rate for the case of emitting or absorbing a phonon respectively,

in any state m_s , it can be expressed as

$$\begin{aligned}\Gamma_{1,m'_s \leftarrow m_s}^{(1),em} &= \frac{2\pi}{\hbar} \sum_l \left| \langle m'_s, n_l + 1 | H_{s-ph} | m_s, n_l \rangle \right|^2 \delta(\hbar\omega_{m'_s m_s} - \hbar\omega_l) \\ &= \frac{2\pi}{\hbar} \sum_l \left| V_{m'_s m_s}^l \right|^2 \left| \langle n_l + 1 | b_l^\dagger + b_l | n_l \rangle \right|^2 \delta(\hbar\omega_{m'_s m_s} - \hbar\omega_l) \\ &= \frac{2\pi}{\hbar} \sum_l \left| V_{m'_s m_s}^l \right|^2 (n_l + 1) \delta(\hbar\omega_{m'_s m_s} - \hbar\omega_l),\end{aligned}\quad (2.32)$$

$$\begin{aligned}\Gamma_{1,m'_s \leftarrow m_s}^{(1),abs} &= \frac{2\pi}{\hbar} \sum_l \left| \langle m'_s, n_l - 1 | H_{s-ph} | m_s, n_l \rangle \right|^2 \delta(\hbar\omega_{m'_s m_s} + \hbar\omega_l) \\ &= \frac{2\pi}{\hbar} \sum_l \left| V_{m'_s m_s}^l \right|^2 n_l \delta(\hbar\omega_{m'_s m_s} + \hbar\omega_l).\end{aligned}\quad (2.33)$$

Since the phonon density increases with the cell-size, the discrete spectrum of spin-phonon coupling coefficients becomes increasingly dense and is therefore better treated in the continuous limit of phonon energy. Taking this limit is done by convolving the discrete spin-phonon coupling coefficients with a Gaussian kernel. The continuous spin-phonon coupling spectrum then defined as

$$J_{m'_s m_s}^{(1)}(\omega) = \int \sum_l \left| V_{m'_s m_s}^l \right|^2 \delta(\hbar\omega' - \hbar\omega_l) \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\hbar\omega' - \hbar\omega)^2}{2\sigma^2}} d\hbar\omega' \quad (2.34)$$

where σ is the standard deviation of the Gaussian and the superscript, as mentioned previously, refers to the order of phonon interaction. The discrete sum over phonon modes in Eqs. (2.32) and (2.33) can be encapsulated in the spectral density, evaluated at the resonant frequency of the spin transition $\hbar\omega_l = \pm \hbar\omega_{m'_s m_s}$, for $E_{m'_s} > E_{m_s}$ and $E_{m'_s} < E_{m_s}$ respectively

$$\begin{aligned}\Gamma_{1,m'_s \leftarrow m_s}^{(1),em}(T) &= \frac{2\pi}{\hbar} (n(|\omega_{m'_s m_s}|) + 1) J_{m'_s m_s}^{(1)}(|\omega_{m'_s m_s}|) \\ \Gamma_{1,m'_s \leftarrow m_s}^{(1),abs}(T) &= \frac{2\pi}{\hbar} n(|\omega_{m'_s m_s}|) J_{m'_s m_s}^{(1)}(|\omega_{m'_s m_s}|)\end{aligned}\quad (2.35)$$

where $n(\omega)$ is the Bose-Einstein occupation number.

Second-order perturbation theory

In second order perturbation theory, Fermi's golden rule is (see Section A.2)

$$\Gamma_{2,f \leftarrow i}^{(1)} = \frac{2\pi}{\hbar} \sum_{l'l'} \left| \langle f | H_{s-ph} G_{\pm} H_{s-ph} | i \rangle \right|^2 \delta(E_f - E_i \pm (\hbar\omega_{l'} \pm \hbar\omega_l)) \quad (2.36)$$

where l and l' refer to two different, independent phonon modes and

$$G_{\pm} = \sum_n \frac{|n\rangle \langle n|}{E_n - E_i \pm \hbar\omega_l}. \quad (2.37)$$

The ZFSs of the NV and CIV centers are relatively small compared to the phonon energies. In order to simplify calculations we neglect the energy of the initial and final spin state in the expression for the transition rate and

$$\begin{aligned}G_{\pm} &\approx \sum_{m''_s} \frac{|m''_s, n''\rangle \langle m''_s, n''|}{\pm \hbar\omega_l} = \frac{1}{\pm \hbar\omega_l} I, \\ \delta(E_{m'_s} - E_{m_s} \pm (\hbar\omega_{l'} \pm \hbar\omega_l)) &\approx \begin{cases} \delta(\hbar\omega_{l'} - \hbar\omega_l) \\ \delta(-\hbar\omega_{l'} + \hbar\omega_l) \\ \delta(\hbar\omega_{l'} + \hbar\omega_l) \\ \delta(-\hbar\omega_{l'} - \hbar\omega_l). \end{cases}\end{aligned}\quad (2.38)$$

One consequence of this approximation is that the two terms with either double absorption or double emission rarely fulfil energy conservation and therefore can be neglected as well. Thus we only consider $\delta(\hbar\omega_{l'} - \hbar\omega_l)$ and can scale the rate by a factor of 2 to account for both the case of first absorbing a phonon and then emitting one and vice versa. This approximation lets us express the total second-order transition rate between states m_s and m'_s as [19]

$$\begin{aligned}\Gamma_{2,m'_s \leftarrow m_s}^{(1)} &= \frac{2\pi}{\hbar} \sum_{ll'} \left| \langle m'_s, n' | H_{s-ph}^{l'} \frac{I}{\pm \hbar\omega_l} H_{s-ph}^l | m_s, n \rangle \right|^2 \delta(\hbar\omega_{l'} - \hbar\omega_l) \\ &= \frac{2\pi}{\hbar} \sum_{ll'} \frac{1}{\hbar^2 \omega_l^2} \left| \langle m'_s, n' | H_{s-ph}^{l'} H_{s-ph}^l | m_s, n \rangle \right|^2 \delta(\hbar\omega_{l'} - \hbar\omega_l) \\ &= \frac{2\pi}{\hbar} \sum_{ll', m''_s} \frac{1}{\hbar^2 \omega_l^2} \left[\left| V_{m'_s m''_s}^l V_{m''_s m_s}^{l'} \right|^2 n_{l'}(n_l + 1) + \left| V_{m'_s m''_s}^{l'} V_{m''_s m_s}^l \right|^2 n_l(n_{l'} + 1) \right] \delta(\hbar\omega_{l'} - \hbar\omega_l).\end{aligned}\quad (2.39)$$

The energy conservation delta forces $l = l'$. In the continuous limit, we get

$$\Gamma_{2,m'_s \leftarrow m_s}^{(1)}(T) = \frac{4\pi}{\hbar} \sum_{m''_s} \int_0^\infty n(\omega)(n(\omega) + 1) \frac{J_{m'_s m''_s}^{(1)}(\omega) J_{m''_s m_s}^{(1)}(\omega)}{\hbar^2 \omega^2} d\hbar\omega \quad (2.40)$$

Two-phonon processes

For second-order phonon interactions, we have the Hamiltonian

$$H_{s-ph} = \sum_{i \in \{x, y, x', y', z\}} \sum_{l \otimes l' \in \Gamma_i} V_i^{ll'} \hat{x}_l \hat{x}_{l'} \hat{F}_i(\hat{\mathbf{S}}) \quad (2.41)$$

which for different phonon pairs give the possible factors $n_l n_{l'}$, $n_l(n_{l'} + 1)$, $(n_l + 1)n_{l'}$, $(n_l + 1)(n_{l'} + 1)$ and the transition rates are split into four different contributions

$$\Gamma_{1,m'_s m_s}^{(2)}(T) = \begin{cases} \frac{4\pi}{\hbar} \sum_{ll'} n_l(n_{l'} + 1) |V_{m'_s m_s}^{ll'}|^2 \delta(\hbar\omega_l - \hbar\omega_{l'}) \\ \frac{2\pi}{\hbar} \sum_{ll'} n_l n_{l'} |V_{m'_s m_s}^{ll'}|^2 \delta(\hbar\omega_{m'_s m_s} + \hbar\omega_l + \hbar\omega_{l'}) \\ \frac{2\pi}{\hbar} \sum_{ll'} (n_l + 1)(n_{l'} + 1) |V_{m'_s m_s}^{ll'}|^2 \delta(\hbar\omega_{m'_s m_s} - \hbar\omega_l - \hbar\omega_{l'}) \end{cases} \quad (2.42)$$

where the cases are for two phonon absorption and emission, double absorption and double emission. For Raman transitions, energy conservation only allows transitions where $\omega_l = \omega_{l'}$ which drops the delta and reduces the expression to a single sum. This is in the continuous limit

$$\Gamma_{1,m'_s m_s}^{(2)}(T) = \begin{cases} \frac{4\pi}{\hbar} \int_0^\infty n(\omega)(n(\omega') + 1) J_{m'_s m_s}^{(2)}(\omega, \omega) d\hbar\omega \\ \frac{2\pi}{\hbar} \int_0^\infty \int_0^\infty n(\omega)n(\omega') J_{m'_s m_s}^{(2)}(\omega, \omega') \delta(\hbar\omega_{m'_s m_s} + \hbar\omega_l + \hbar\omega_{l'}) d\hbar\omega d\hbar\omega' \\ \frac{2\pi}{\hbar} \int_0^\infty \int_0^\infty (n(\omega) + 1)(n(\omega') + 1) J_{m'_s m_s}^{(2)}(\omega, \omega') \delta(\hbar\omega_{m'_s m_s} - \hbar\omega_l - \hbar\omega_{l'}) d\hbar\omega d\hbar\omega' \end{cases} \quad (2.43)$$

where the temperature dependance comes from the Bose-Einstein distributions.

$$J_{m'_s m_s}^{(2)}(\omega, \omega') = \int \sum_{l'l'} |V_{m'_s m_s}^{l'l'}|^2 \delta(\hbar\nu - \hbar\omega_l) \delta(\hbar\nu' - \hbar\omega_{l'}) \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\hbar\nu - \hbar\omega)^2 + (\hbar\nu' - \hbar\omega')^2}{2\sigma^2}} d\hbar\nu d\hbar\nu' \quad (2.44)$$

In Figure 2.8, we see all possible second-order transitions for when $E_{m'_s} > E_{m_s}$.

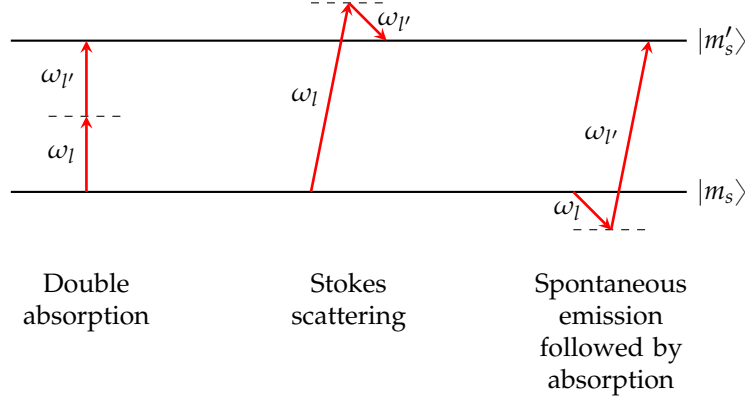


Figure 2.8: Possible second order transitions between states $|m'_s\rangle$ and $|m_s\rangle$ in the case of $E_{m'_s} > E_{m_s}$. Figure inspired by Ref. [36].

2.6 Density functional theory

Density functional theory (DFT) [37] is a computational method in quantum physics and chemistry for finding the ground-state electronic structure of many-body systems like atoms, molecules or solids, using the electron density $n(\mathbf{r})$, rather than the many-body wavefunction. It is based on the Hohenberg-Kohn theorems [38] which state

Theorem 2.6.1: Hohenberg-Kohn

1. For a many-body system with an external potential V_{ext} , the external potential is uniquely described by the charge density $n(\mathbf{r})$ up to an additive constant, and thus also the total energy E .
2. The universal energy functional $E[n]$ has its minimum when the charge density is that of the true ground state $n_0(\mathbf{r})$, given that the number of particles stays constant.

A consequence of the Hohenberg-Kohn theorems is that the full Hamiltonian can be described by the ground-state electron density $n(\mathbf{r})$. Working with the electron density instead of the many-body wavefunction $\Psi(\mathbf{r})$ is more computationally efficient; the number of variables scales with $3n$ when working with wavefunctions with n number of particles while the density has three variables regardless of the number of particles.

Further, Kohn and Sham have shown that there exists a system of non-interacting particles described by a potential $V_{eff}(\mathbf{r})$ with the same electron density as the interacting system [39]. Solving the Kohn-Sham equations, i.e., a collection of non-interacting single-particle equations, thus yields the exact ground-state density and total energy of the interacting system, even though the Kohn-Sham orbitals are those of a non-interacting system. This means that the exact ground state energy of a system of interacting particles is given by

$$E[n] = T_s[n] + \int V_{ext}(\mathbf{r})n(\mathbf{r})d\mathbf{r} + \frac{1}{2} \iint \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' + E_{xc}[n] \quad (2.45)$$

where $T_s[n]$ is the kinetic energy for a non-interacting system and $E_{xc}[n]$ is the exchange-correlation energy functional containing all many-body interactions. The exact exchange-correlation energy functional is not known and is therefore approximated using different functionals like Perdew–Burke–Ernzerhof (PBE) [40] or Heyd–Scuseria–Ernzerhof (HSE) [41].

Minimising the energy functional $E[n]$ with respect to the wavefunctions of the non-interacting system gives the Kohn-Sham equations

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{eff}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}) \quad (2.46)$$

where the effective potential $V_{eff}(\mathbf{r})$ for the non-interacting system is

$$V_{eff}(\mathbf{r}) = V_{ext}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{xc}[n]}{\delta n(\mathbf{r})}. \quad (2.47)$$

The wavefunctions ψ_i are then used to find a new charge density $n'(\mathbf{r})$

$$n'(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2. \quad (2.48)$$

where the sum runs over the occupied Kohn-Sham orbitals containing the N particles in the system. This process is iterated until convergence is achieved, usually by looking at the difference in total energy or charge density between iterations. In spin-DFT, one separates the electron density into a spin-up n_α and spin-down n_β and describe the total density as $n_{tot} = n_\alpha + n_\beta$ and the magnetic spin density $n_{mag} = n_\alpha - n_\beta$. Integrating the density n_{mag} over the volume of the crystal yields the magnetization of the crystal.

Spin contamination

In open-shell systems, DFT typically utilize an unrestricted Kohn-Sham formalism to evaluate the spin polarization. This allows the spatial parts of the spin up α and spin down β wave function to differ, in order to reach a lower energy state. The total wave function of the many body state is then described by a single slater determinant

$$\Psi_{KS}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \cdots & \psi_1(\mathbf{x}_N) \\ \vdots & \ddots & \vdots \\ \psi_N(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix} \quad (2.49)$$

where $\mathbf{x}_i$ refer to both the spatial and spin coordinates. While relaxation of the spatial spin wave function is energetically favourable it leads to Ψ_{KS} no longer being an eigenfunction of the total spin operator S^2 . For an exact pure spin state, the expectation value of S^2 must satisfy

$$\langle S^2 \rangle = S(S+1). \quad (2.50)$$

However, evaluating S^2 projected onto Ψ_{KS} yields

$$\langle \Psi_{KS} | S^2 | \Psi_{KS} \rangle = S(S+1) + N_\beta - \sum_i^{N_\alpha} \sum_j^{N_\beta} \langle \psi_i^\alpha | \psi_j^\beta \rangle \quad (2.51)$$

where $\langle \psi_i^\alpha | \psi_j^\beta \rangle$ represents the spatial overlap between the α and β spin orbitals. In a restricted formalism, the sum of the overlap would evaluate to N_β and thus result in a pure spin value of $S(S+1)$. In the unrestricted formalism, the spatial overlap will decrease as the spin density relaxes and $\sum \langle \psi_i^\alpha | \psi_j^\beta \rangle < N_\beta$. We can then denote the spin contamination as [42]

$$C_s = N_\beta - \sum_i^{N_\alpha} \sum_j^{N_\beta} \langle \psi_i^\alpha | \psi_j^\beta \rangle. \quad (2.52)$$

Only the unpaired spins are relevant in a dipole-dipole interaction but the spin contamination will affect the $\mathbf{D}$ -tensor due to non-physical contributions from the bulk orbitals. In order to mitigate this error in ZFS calculations, Biktagirov et al. [25] suggest a method of calculating the ZFS for the spin state $m_s = S$ and all possible configurations of $m_s = S - 1$. From our expression for the ZFS tensor in Eq. (2.17) we can define a non normalized tensor $\mathbf{d}$ as

$$\tilde{\mathbf{D}} = \frac{\mathbf{d}}{S(S-1/2)}. \quad (2.53)$$

The corrected ZFS tensor is then calculated from the difference between the ZFS of the highest spin state $m_s = S$ and the second highest state $m_s = S - 1$

$$\tilde{\mathbf{D}}_{corrected} = \frac{\mathbf{d}_{m_s=S} - \mathbf{d}_{m_s=S-1}}{2S-1}. \quad (2.54)$$

To reduce the computational cost of the spin correction method, an approximated method of Biktagirov et al. is used in this thesis, motivated from the supplementary materials of Ref. [25]. This approximated method calculates the ZFS tensor using only the unpaired electrons and relates it to the full spin corrected tensor via a scaling factor:

$$\tilde{\mathbf{D}}_{corrected} \approx \frac{3}{2} \tilde{\mathbf{D}}^{I,I} \quad (2.55)$$

where $\tilde{\mathbf{D}}^{I,I}$ denotes the ZFS tensor of the unpaired electrons. For spin-1 systems, the scaling factor is 3/2.

2.7 T_1 Relaxation time

The theoretical T_1 time is defined as the time it takes for the longitudinal component M_z of the magnetization vector to recover its thermodynamic equilibrium, governed by the equation

$$M_z(t) = M_{z,eq} - [M_{z,eq} - M_z(0)]e^{-\frac{t}{T_1}} \quad (2.56)$$

where $M_{z,eq}$ is the equilibrium value, $M_z \propto \int n_{mag} dV$. For a multi-state system, the T_1 relaxation time can be extracted by solving the system of ODEs describing the population change of the different states

$$\frac{d\mathbf{p}}{dt} = \mathbf{A}\mathbf{p}(t) \quad (2.57)$$

where $\mathbf{p}(t)$ and $\mathbf{A}$ is the population vector and the transition rate matrix respectively

$$\mathbf{p}(t) = (p_1(t) \quad \cdots \quad p_N(t))^T$$

$$\mathbf{A} = \begin{pmatrix} \Gamma_{1,1} & \cdots & \Gamma_{1,N} \\ \vdots & \ddots & \vdots \\ \Gamma_{N,1} & \cdots & \Gamma_{N,N} \end{pmatrix}.$$

For the NV center three state system, this becomes

$$\frac{d\mathbf{p}}{dt} = \begin{pmatrix} -\Gamma_{1 \rightarrow 0} - \Gamma_{1 \rightarrow -1} & \Gamma_{0 \rightarrow 1} & \Gamma_{-1 \rightarrow 1} \\ \Gamma_{1 \rightarrow 0} & -\Gamma_{0 \rightarrow 1} - \Gamma_{0 \rightarrow -1} & \Gamma_{-1 \rightarrow 0} \\ \Gamma_{1 \rightarrow -1} & \Gamma_{0 \rightarrow -1} & -\Gamma_{-1 \rightarrow 0} - \Gamma_{-1 \rightarrow 1} \end{pmatrix} \begin{pmatrix} p_{m_s=1}(t) \\ p_{m_s=0}(t) \\ p_{m_s=-1}(t) \end{pmatrix} \quad (2.58)$$

which has an analytical solution

$$\mathbf{p}(t) = \sum_i c_i \mathbf{u}_i e^{\lambda_i t} \quad (2.59)$$

where $\mathbf{u}_i$ and λ_i is the eigenvector and eigenvalue of mode i . The coefficients c_i are determined by boundary conditions. Because of probability conservation $\sum_j A_{ij} = 0$, one eigenvalue is always zero, corresponding to the steady state population distribution. All other eigenvalues are negative, describing the population decay of the non-equilibrium states. This gives the solution

$$\mathbf{p}(t) = c_0 \mathbf{u}_0 + \sum_{i>0} c_i \mathbf{u}_i e^{\lambda_i t} \quad (2.60)$$

where $\mathbf{u}_0$ is the steady state solution. An issue arises if the solution is evenly distributed across multiple modes as the T_1 time no longer can be extracted from a single eigenvalue. To mitigate this issue, an exponential fit can be done on the specific solution $\mathbf{p}_s(t)$ of Eq. (2.57) with a given boundary condition $\mathbf{p}(t=0)$.

This section presents the computational framework used to calculate the transition rates and T_1 relaxation time. First, the computational details of DFT and ZFS calculations are presented. A symmetry analysis of phonon modes is performed using a home-made script. The phonon data was made available through previous computations in the research group for materials design and informatics at LiU, where the modes were obtained via the method of finite displacements [21]. As the number of E -modes constitute approximately 2/3 of all phonon modes, the symmetry analysis reduces the number of DFT simulations by a factor of $\sim 1/3$. Further, the relaxation rate calculation from spin-phonon coupling coefficients is presented. Figure 3.1 shows a flowchart of the full computational process to derive the transition rates.

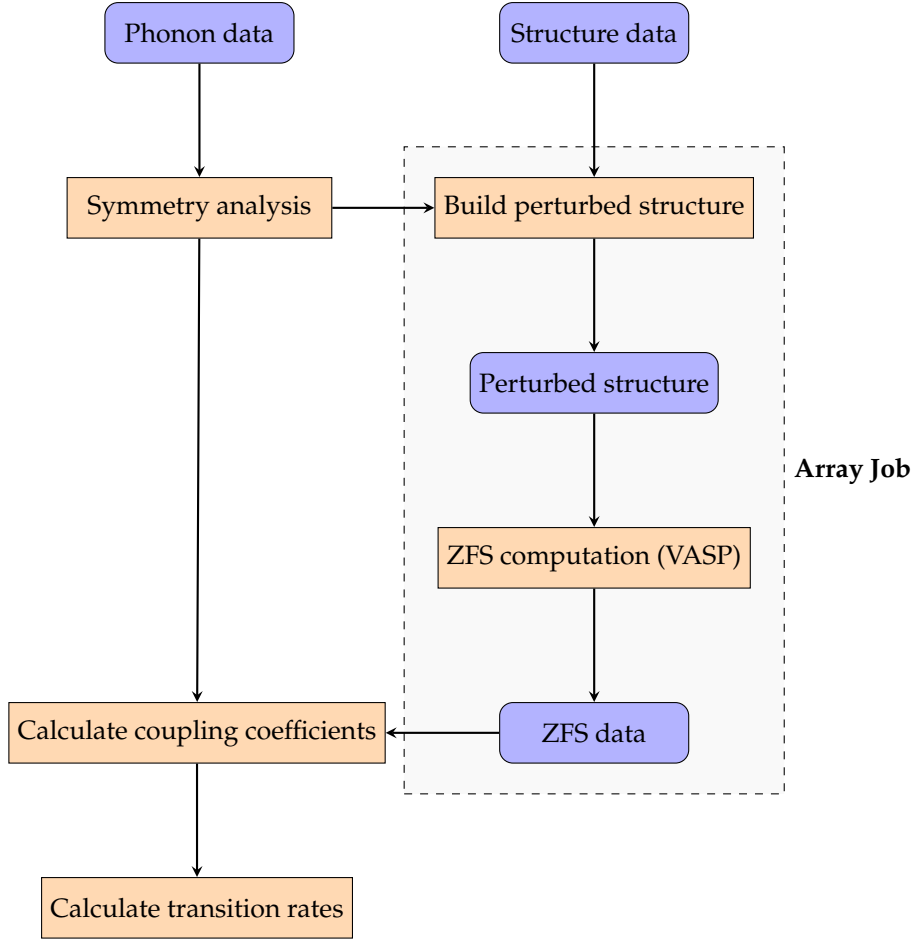


Figure 3.1: Flowchart for the process of calculating the relaxation rate, from input of crystal structure data to relaxation rate calculation from ZFS coefficients.

3.1 Computational details

The studied systems include a 512 and 64 atom supercell NV center defect, and a 128 atom supercell CIV center. The ZFS is calculated using DFT simulations run in Vienna Ab initio

Simulation Package (VASP) [43–48]. These calculations are split into two separate stages; first, the relaxed crystal structure is simulated. From the obtained solution, the occupation of spin orbitals is used to initialize the calculation of the ZFS-tensor on the relaxed structure. The electronic structure simulations are performed using the PBE functional projector augmented wave (PAW) method with a cutoff energy of 600 eV. The convergence criteria for the NV center is 1×10^{-7} eV and 1×10^{-6} eV in total energy, and 1×10^{-5} eVÅ⁻¹ and 1×10^{-3} eVÅ⁻¹ for forces in ionic relaxation, for the 64 and 512 atom cell respectively. For the CIV center simulations, the criteria is 1×10^{-10} eV in total energy and 1×10^{-7} eVÅ⁻¹ for forces in ionic relaxation. Symmetry analysis and constraints are applied when simulating the ground state relaxed structure for computational efficiency.

Two methods were used when calculating the ZFS. The first method uses all occupied orbitals. The second method uses only the orbitals of the unpaired electrons, the outer most ground state orbitals in the spin up channel. For this method, a modified version of VASP was used in order to fix the spin occupation in the ZFS calculation.

Symmetry analysis of phonon modes

A symmetry analysis of the phonon modes is performed to reduce the number of perturbed structure DFT simulations and to categorize the coupling coefficients to the different symmetry-adapted terms of the Hamiltonian in Eq. (2.26). The symmetry analysis applies the operations C_3 and σ_v of the C_{3v} point group to the eigenvector of each phonon mode, computes the overlap with the original vector, thus mapping the character. Before applying the symmetry operations, the defect position in the supercell is found and a change of basis is done to align the symmetry axis of the defect to the z-axis. *E*-mode basis pairs are identified using their degeneracy as the matching criteria with an energy threshold of 1×10^{-2} meV.

Perturbed ZFS calculation

The ZFS calculation needs the wavefunctions of the perturbed structures. These are found in VASP by constraining the relaxation of ionic positions, with the initial wavefunction taken from the relaxed structure. The perturbed structure DFT simulations applies no symmetry constraints for optimizing electron density calculations to minimize the risk of the VASP symmetry analyser to incorrectly identify the perturbed structure as C_{3v} , in cases of symmetry-breaking *E*-modes. The perturbed structures are simulated at two degrees of phonon perturbation, one using single phonon modes and the second using two phonon modes. The perturbation is applied as a shift of ionic positions according to each eigenvector of the treated phonon mode with a select amplitude, which was selected based on a convergence test of multiple amplitudes on the 64 atom NV center, of $0.1 \text{ Å}\sqrt{\text{a.m.u}}$ for the NV center simulations and $0.025 \text{ Å}\sqrt{\text{a.m.u}}$ for the CIV center simulations. The NV center perturbation of $0.1 \text{ Å}\sqrt{\text{a.m.u}}$ is chosen to better compare the results to Ref. [19]. The perturbed structures are created using the modules `pymatgen.core.structure` and `pymatgen.io.vasp.input` of the Python library Python Materials Genomics (pymatgen) [49]. After electron density relaxation is converged, the ZFS is computed using the aforementioned methods. From the ZFS-tensors computed from the perturbed crystal structures, the first and second order derivatives of the ZFS-tensor with respect to ionic displacements are calculated as

$$\left\{ \begin{array}{l} \frac{d\tilde{\mathbf{D}}}{dQ_I} \approx \frac{\tilde{\mathbf{D}}(Q_I) - \tilde{\mathbf{D}}(\mathbf{Q} = \mathbf{0})}{Q_I} \end{array} \right. \quad (3.1a)$$

$$\left\{ \begin{array}{l} \frac{d\tilde{\mathbf{D}}}{dQ_I dQ_{I'}} \approx \frac{\tilde{\mathbf{D}}(Q_I, Q_{I'}) - \tilde{\mathbf{D}}(Q_I) - \tilde{\mathbf{D}}(Q_{I'}) + \tilde{\mathbf{D}}(\mathbf{Q} = \mathbf{0})}{Q_I Q_{I'}}. \end{array} \right. \quad (3.1b)$$

Because of energy conservation in Fermi's golden rule in Eqs. (2.30) and (2.36), we only need to simulate two-phonon perturbations with pairs of the same phonon mode, as pairs of modes with large differences in energy would break this energy conservation. From the symmetry analysis of the interaction hamiltonian, only one of the degenerate E_x and E_y modes need to be simulated since they give the same contribution to the coupling coefficients and can thus be compensated in post processing of the simulation data.

3.2 Calculating the relaxation rate

The transition rate is calculated according to Eqs. (2.30), (2.36) and (2.43). Instead of evaluating the continuous integrals, the calculations are performed as discrete sums over a fine energy grid with spacing ΔE . The energy conserving Dirac delta is approximated by a Gaussian of standard deviation of $\sigma = 1 \text{ meV}$, so that each transition rate becomes a convolution of the spectral density with the appropriate thermal occupation factors.

For the first-order (direct) processes, the spin-phonon coupling coefficients $V_{0\pm}^I$ and V_{+-}^I are squared and smeared with a chosen standard deviation $\sigma = 7.5 \text{ meV}$ to obtain the one-dimensional spectral density functions $J_{0\pm}^{(1)}(\omega)$ and $J_{+-}^{(1)}(\omega)$ on a grid with a resolution of $\sim 0.01 \text{ meV}$. The transition rates for a transition $m_s \rightarrow m'_s$ are then

$$\Gamma_{1,m_s \rightarrow m'_s}^{(1)} = \frac{2\pi}{\hbar} \Delta E \sum_i J_{m_s m'_s}(\omega_i) \begin{cases} n(\omega_i) & \text{(absorption)} \\ n(\omega_i) + 1 & \text{(emission)} \end{cases} \times G(\omega_i - \Delta_{m_s m'_s}), \quad (3.2)$$

where ΔE is the grid spacing, $n(\omega) = (e^{\hbar\omega/k_B T} - 1)^{-1}$ is the Bose-Einstein distribution, $\Delta_{m_s m'_s}$ is the energy difference between the spin sublevels, and G denotes the Gaussian approximation of the delta function. Only phonon energies above 0.1 meV are considered to avoid numerical overflow in the evaluation of $n(\omega)$.

Second-order (Raman) processes arise from two virtual phonons that connect the initial and final spin states m_s and m'_s through an intermediate state m''_s . They are computed using the same first order spectral densities. The rate for a transition $m_s \rightarrow m'_s$ is obtained by summing over all possible intermediate states

$$\Gamma_{2,m_s \rightarrow m'_s}^{(2)} = \frac{4\pi}{\hbar} \Delta E \sum_{m''_s} \sum_i \frac{J_{m'_s m''_s}(\omega_i) J_{m''_s m_s}(\omega_i)}{(\hbar\omega_i)^2} n(\omega_i) (n(\omega_i) + 1). \quad (3.3)$$

Two-phonon processes driven by the second-order derivative of the ZFS (i.e. the coefficients $V_{0\pm}^{II}$, V_{+-}^{II} , and V_{00}^{II}) require a two-dimensional spectral density $J_{m_s m'_s}^{(2)}(\omega_1, \omega_2)$. This 2D spectral density is constructed by smearing the squared coupling coefficients with a 2D Gaussian function on a grid (ω_1, ω_2) with a resolution of 0.5 meV and the same Gaussian width of 7.5 meV as in the first order spectral functions. For a given spin transition $m_s \rightarrow m'_s$ the rate is given by a sum over four different combinations of phonon absorption and emission

$$\Gamma_{1,m_s \rightarrow m'_s}^{(2)} = \frac{2\pi}{\hbar} (\Delta E)^2 \sum_{s,s'=\pm 1} \sum_{i,j} J_{m_s m'_s}(\omega_i, \omega_j) \mathcal{B}_{ss'}(\omega_i, \omega_j) G(s\omega_i + s'\omega_j - \Delta_{m_s m'_s}) \quad (3.4)$$

where the thermal factors $\mathcal{B}_{ss'}$ are given by appropriate products of $n(\omega)$ and $n(\omega) + 1$ corresponding to the process (abs-abs, abs-em, em-abs, and em-em).

Finally, the total spin relaxation rates are obtained by summing all contributions that connect the relevant pairs of spin sublevels. The single quantum rate $\Gamma_{0 \leftrightarrow \pm 1}$ includes all transitions between states $m_s = 0$ and $m_s = \pm 1$ in both directions. The double quantum transition rate $\Gamma_{+1 \leftrightarrow -1}$ combines both directions of ± 1 transitions. For each temperature T , the rates are computed using the framework described above and stored for later analysis.

3.3 Calculation of the T_1 relaxation time

An initial guess of the spin-lattice relaxation time T_1 is computed by evaluating the eigenvalues of the transition matrix $\mathbf{A}$ governing Eq. (2.57) as

$$T_1 = -\frac{1}{\lambda}. \quad (3.5)$$

where λ is the eigenvalue corresponding to the slowest decaying eigenmode. The matrix $\mathbf{A}$ is constructed at each temperature from the total transition rates (first-order, second-order Raman, and two-phonon contributions) that have been calculated on a discrete temperature grid.

For a given temperature T , the matrix $\mathbf{A}(T)$ is constructed according to Eq. (2.57) with off-diagonal elements $A_{ij} = \Gamma_{j \rightarrow i}$ and diagonal elements $A_{ii} = -\sum_{j \neq i} \Gamma_{i \rightarrow j}$. The matrix is diagonalized numerically, yielding the eigenvalues λ_i and eigenvectors $\mathbf{u}_i$.

The slowest relaxation mode is identified as the one with the smallest absolute value among those that are not essentially zero. To avoid numerical noise, any eigenvalue $< 10^{-9} \text{s}^{-1}$ is excluded. The largest remaining eigenvalue λ_{max} is then used to obtain the T_1 time according to Eq. (3.5). This procedure is repeated for every temperature on the chosen interval producing the temperature dependent relaxation time $T_1(T)$.

In cases where the initial condition populates a single eigen mode, the analytical solution reduces to a pure exponential decay with time constant $-1/\lambda_{max}$. If the initial condition does not populate a single eigenmode the population decay is described by a sum of exponentials and the T_1 time can not be extracted from a single eigenvalue. The T_1 time of the polarized initial state $p_{m_s}(t)$ is therefore extracted by fitting a single exponential optimized over T_1 and an additive constant c to the simulated population evolution

$$\min_{T_1, c} \left| \left(e^{-t/T_1} + c \right) - p_{m_s}(t) \right|. \quad (3.6)$$

The spin-phonon coupling coefficients are calculated for the NV center in diamond and the CIV center in SiC. Multiple amplitudes ($0.1 \text{ \AA}\sqrt{\text{a.m.u}}$, $0.05 \text{ \AA}\sqrt{\text{a.m.u}}$ and $0.025 \text{ \AA}\sqrt{\text{a.m.u}}$) of phonon perturbations are tested on the 64 atom NV center and it was concluded that smaller perturbation scales did not affect the coupling coefficients substantially. Therefore, an amplitude of $0.1 \text{ \AA}\sqrt{\text{a.m.u}}$ was picked for the NV center in order to compare the results with previous work. However, an amplitude of $0.025 \text{ \AA}\sqrt{\text{a.m.u}}$ was picked for the CIV center. These results are presented below. The IPR is also compared to the coupling coefficients and are presented in Figures (4.3), (4.4) and (4.11) for each defect.

4.1 NV center in diamond

Two supercell sizes, 64 and 512 atoms, of the NV center were simulated and the results of each are presented in the sections below. The perturbation applied to the atoms was $0.1 \text{ \AA}\sqrt{\text{a.m.u}}$ scaled with the eigenvectors of the phonon modes. The results consist of spin-phonon coupling coefficients, spectral functions, IPR and transition rates for each supercell size and spin occupation method.

Coupling coefficients and Spectral functions

The coupling coefficients and spectral function for first and second order phonons with spin contamination are presented in Figures (4.1a) and (4.1c) and spin decontaminated results are presented in Figures (4.1b) and (4.1d). The spectral functions of single and double quantum transitions in the NV center display the same two peaks at 62.5 meV and 165 meV as stated by previous research [19, 36], see Figure 4.1a. There are two distinct bands of high spin-phonon coupling in the first-order interactions in both supercells and in the two-phonon coupling for the 64 atom supercell. The spin contaminated coupling coefficients of the 512 atom two-phonon perturbation does not display the same structure as the 64 atom cell. The 64 atom supercell has fewer phonons, especially noticeable in the low energy region $\hbar\omega < 50 \text{ meV}$. Phonon modes of E -symmetry dominate the first-order spin-phonon coupling in both the 64 and 512 atom supercell. This effect is significantly reduced for the first-order spin-corrected ZFS in both supercells. The coupling strength of A_1 modes is almost unaffected by the spin contamination correction. In the second-order coupling, A_1 modes are seen to dominate and the coupling strength is especially high in the optical phonon band $\hbar\omega > 125 \text{ meV}$. The second-order phonon interaction for the spin corrected results is seen to be much stronger than the spin contaminated results, compare Figures (4.1c) and (4.2c) with Figures (4.1d) and (4.2d). This effect is opposite to what is seen in the first-order interactions. Phonons with low energy drive direct transitions and double absorption/emission processes. The spin-phonon coupling for low energy phonons are very weak in both the 64 and 512 atom supercell, especially at the ZFS energy of $2.88 \text{ GHz} \approx 0.012 \text{ meV}$.

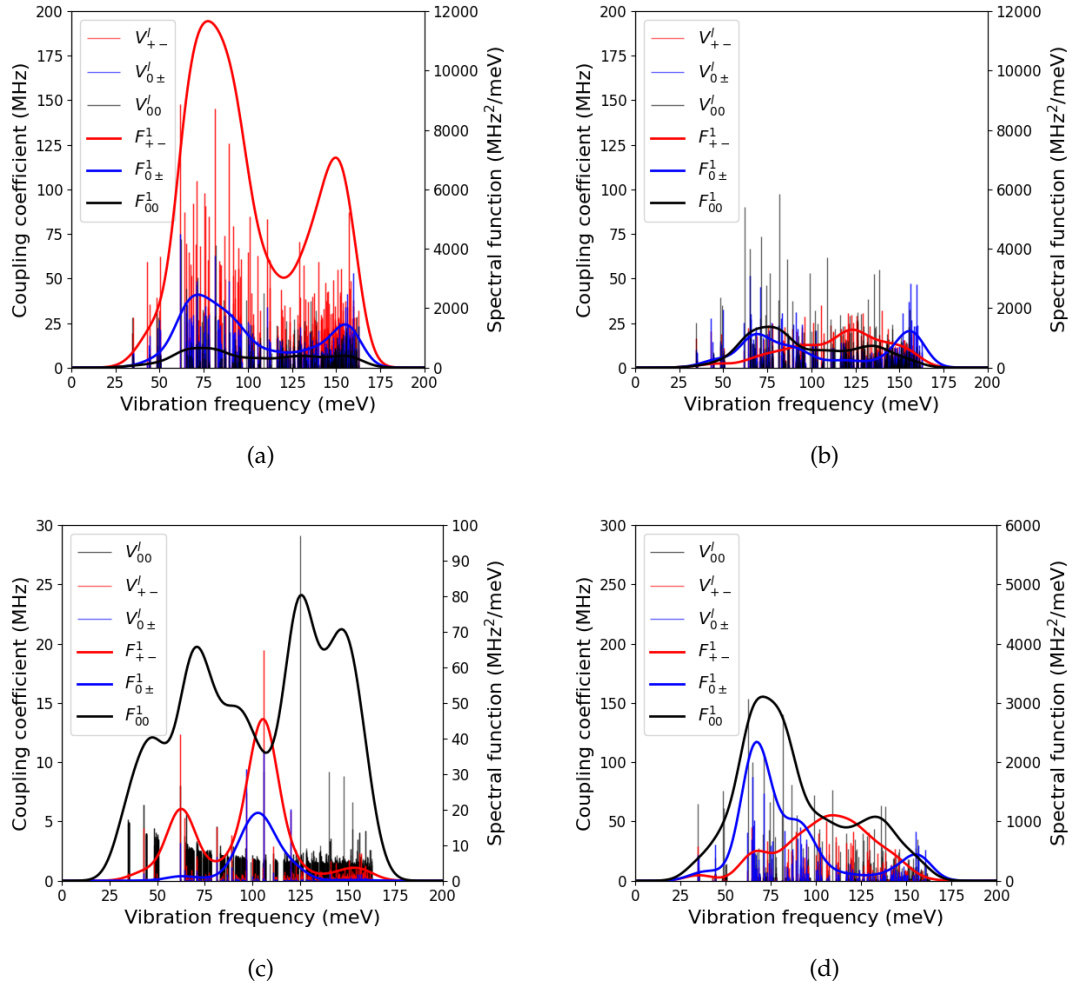


Figure 4.1: The coupling coefficients and spectral function for first and second order phonons for 512 atom supercell. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

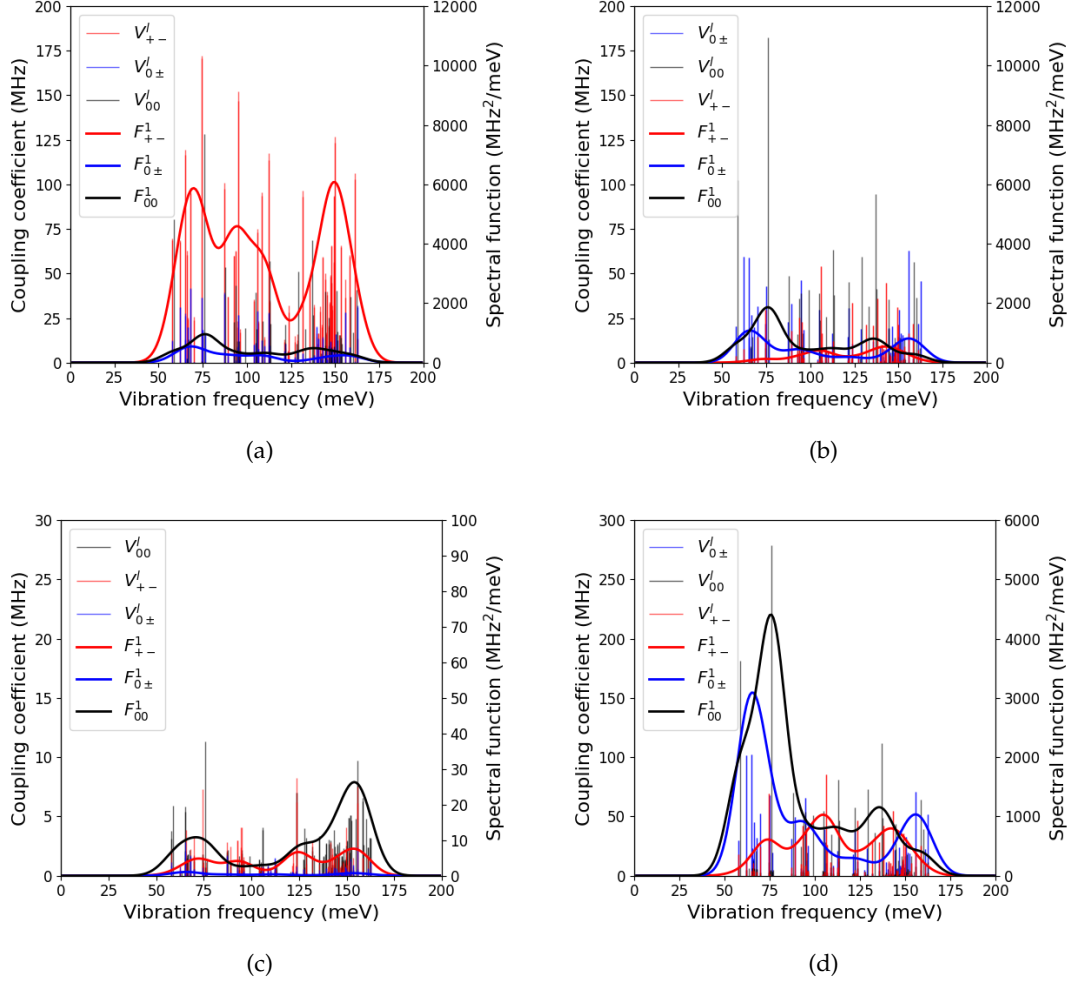


Figure 4.2: The coupling coefficients and spectral function for first and second order phonons for 64 atom supercell. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

Phonon mode localization vs coupling strength

There are some highly localized modes, at 65 meV. However, not all of them have a high spin-phonon coupling, see Figure 4.4c. The localized modes with high coupling strength are also noted in the supplementary materials of Ref. [19]. There are, on the other hand, multiple bulk modes with low IPR which have a high spin-phonon coupling, see Figure 4.3. We therefore note that there is no clear correlation between phonon localization and spin-phonon coupling strength.

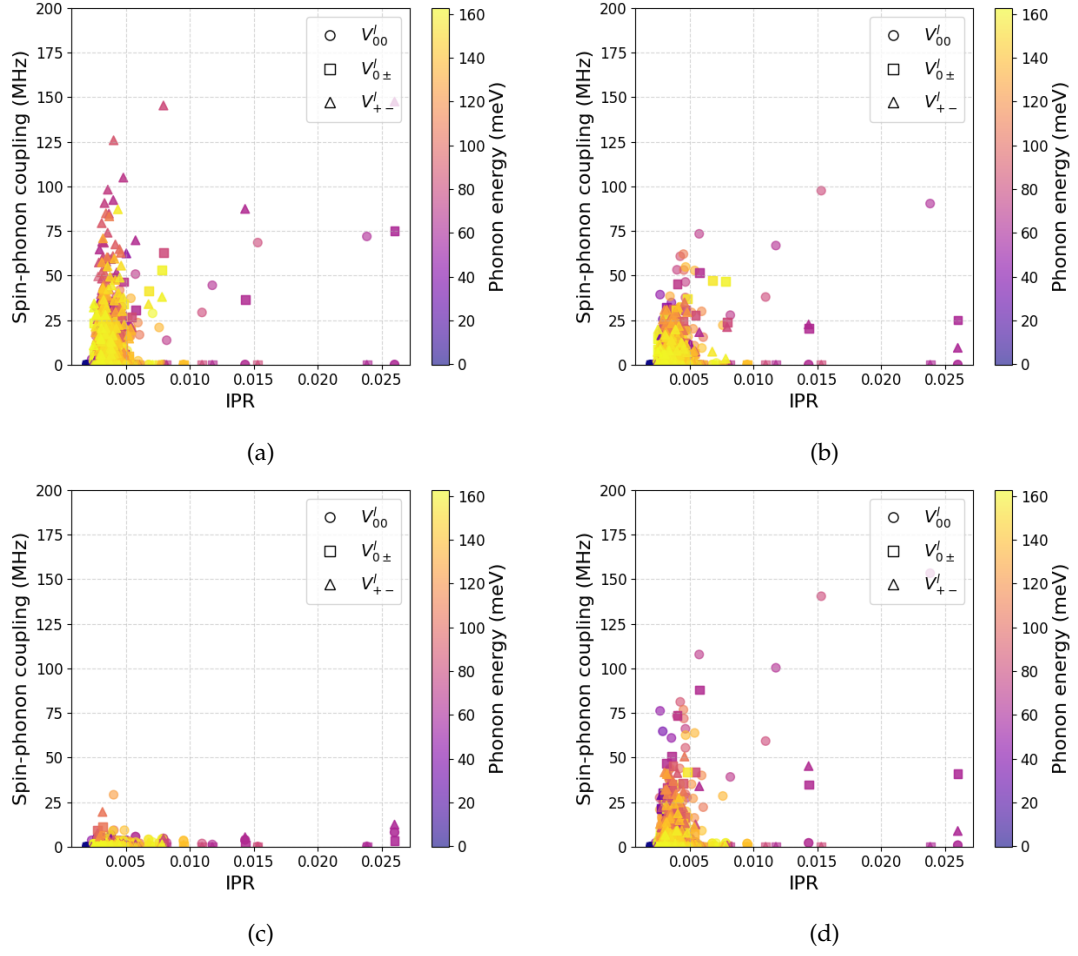


Figure 4.3: The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 512 atom supercell NV center. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

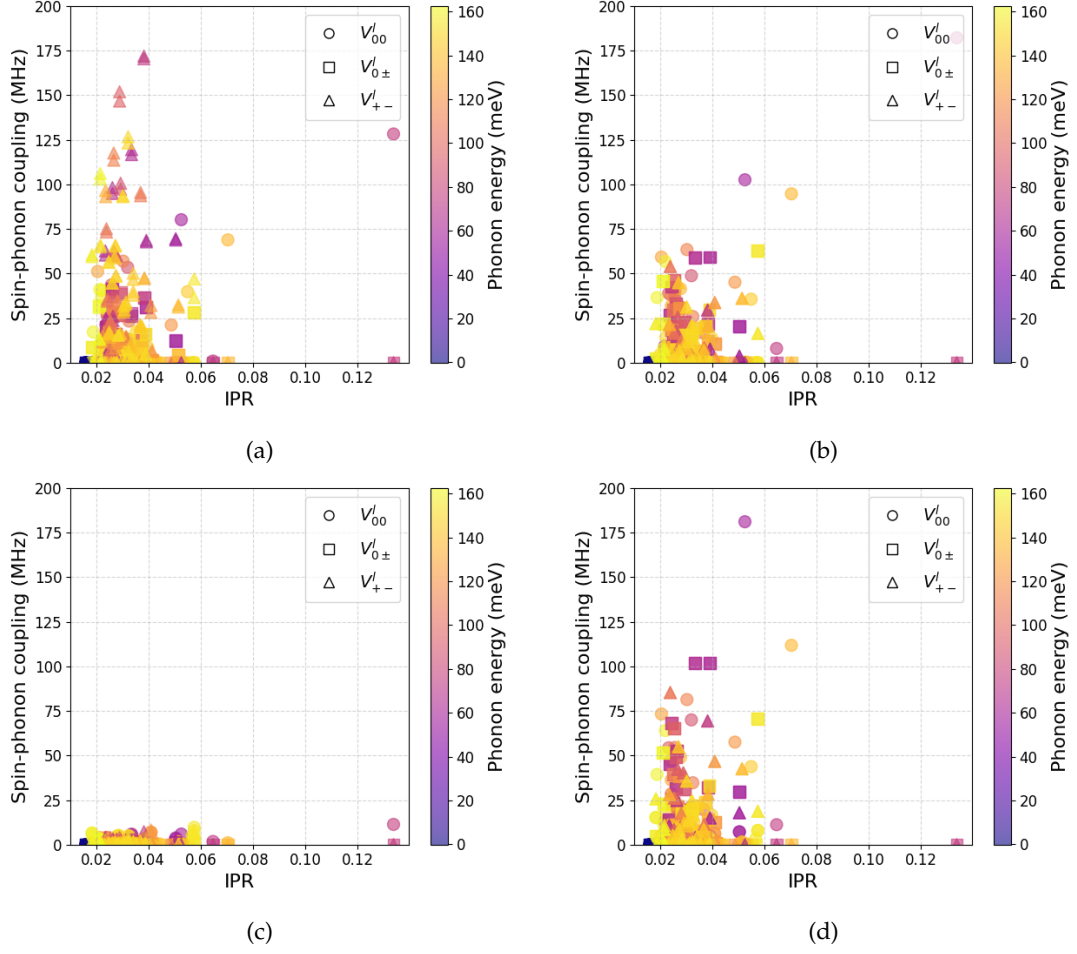


Figure 4.4: The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 64 atom supercell NV center. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

Transition rates

The different transition rates for the NV center dominate different temperature regions, see Figures (4.5) and (4.6). For temperatures below 20 K, the first-order rates dominate with an amplitude of 100 nHz. After 30 K, two-phonon processes begin to rise and it is clear that Raman processes dominate at higher temperatures, as seen in Figure 4.5, which is in agreement with Ref. [19, 36]. The rates from spin corrected coefficients are not aligned with experimental values of the relaxation rate [14, 19, 50] which state that double quantum transitions should outpace single quantum transitions, in contrast to the reverse ordering in Figures (4.5b) and (4.5d). The 64 atom cell has no noticeable contributions from direct transitions, as seen in Figure 4.6.

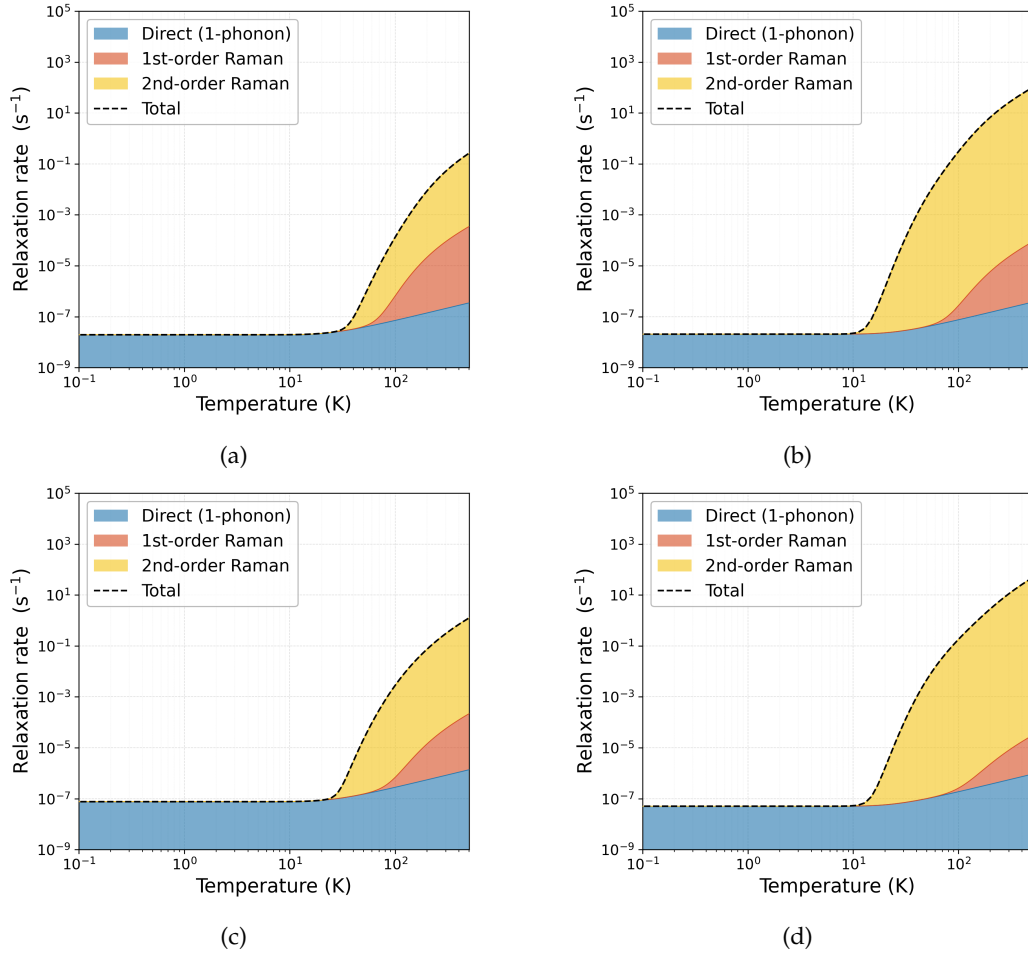


Figure 4.5: Transition rates for a 512 atom supercell size NV center. **(a)** Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and **(b)** with spin corrected ZFS. **(c)** Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and **(d)** with spin corrected ZFS.

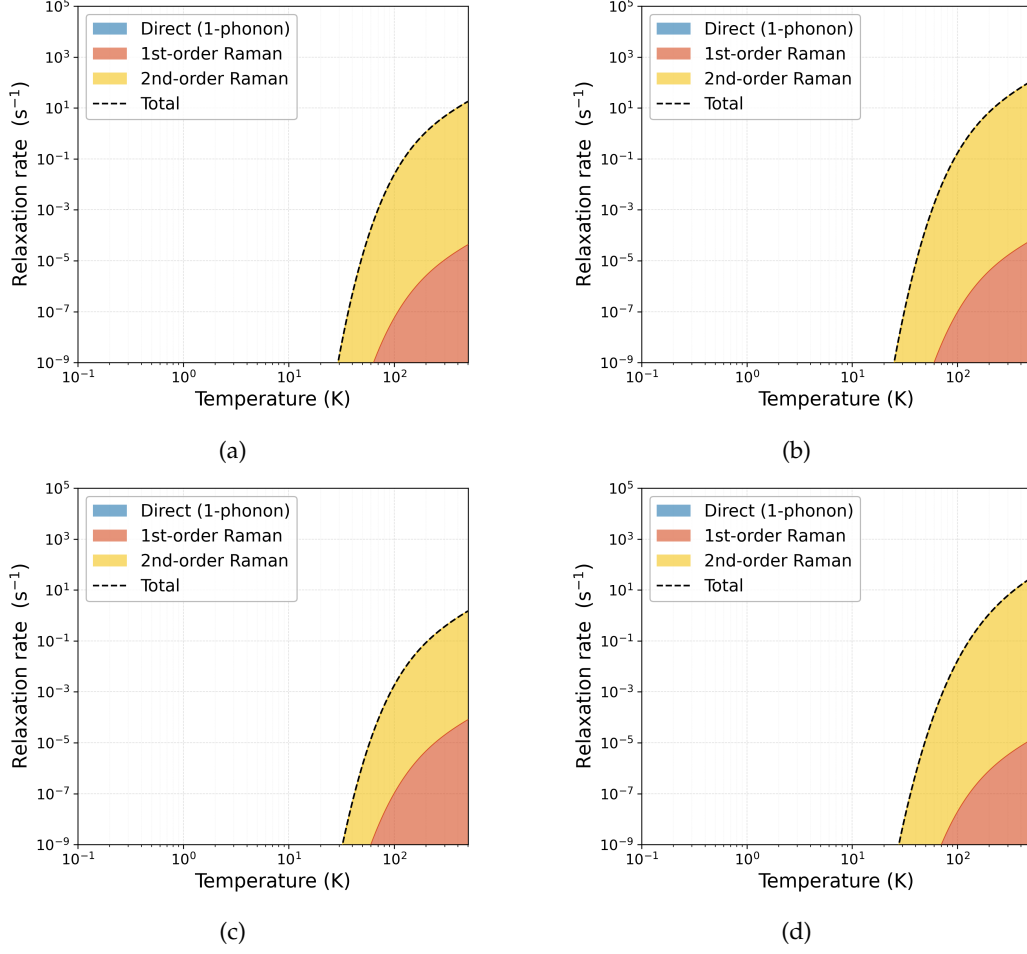


Figure 4.6: Transition rates for a 64 atom supercell size NV center. **(a)** Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and **(b)** with spin corrected ZFS. **(c)** Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and **(d)** with spin corrected ZFS.

T_1 relaxation time

We see that the model is inconsistent in the low temperature region $T < 50$ K, see Figures (4.7) to (4.9). It is many orders of magnitude larger than experimental measurements [51]. The room temperature results are in the expected range of a couple of milliseconds [7, 52] for both supercells in the spin corrected data. From Figures (4.6) and (4.7), we conclude that a small supercell of 64 atoms is sufficient for calculating a qualitative estimation of the T_1 time, in the high temperature region, for the NV center as they are well aligned with those of Cambria et al. [19]. The room temperature T_1 times for the three polarized states $m_s = 0, \pm 1$ are seen in Table 4.1.

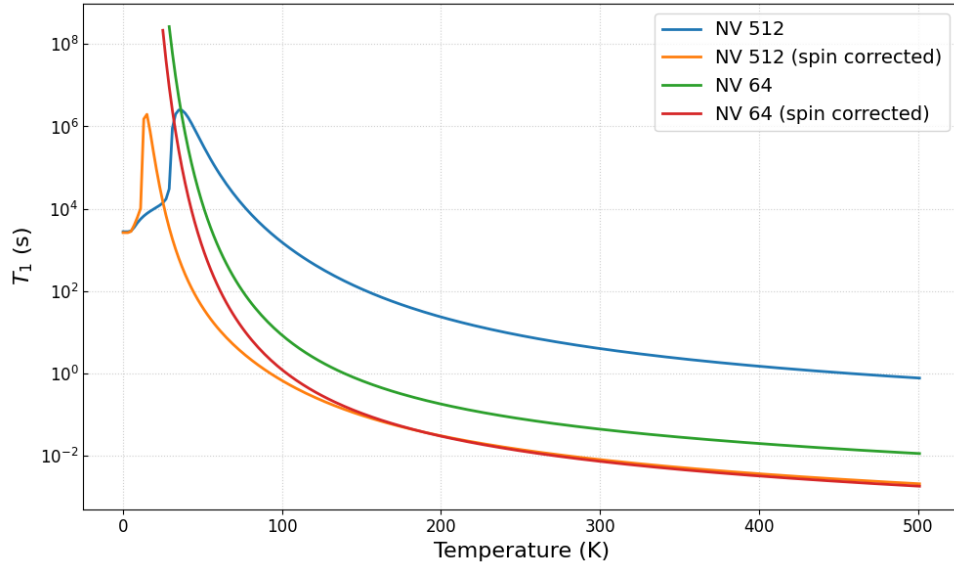


Figure 4.7: The T_1 relaxation time, for the $m_s = 0$ polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.

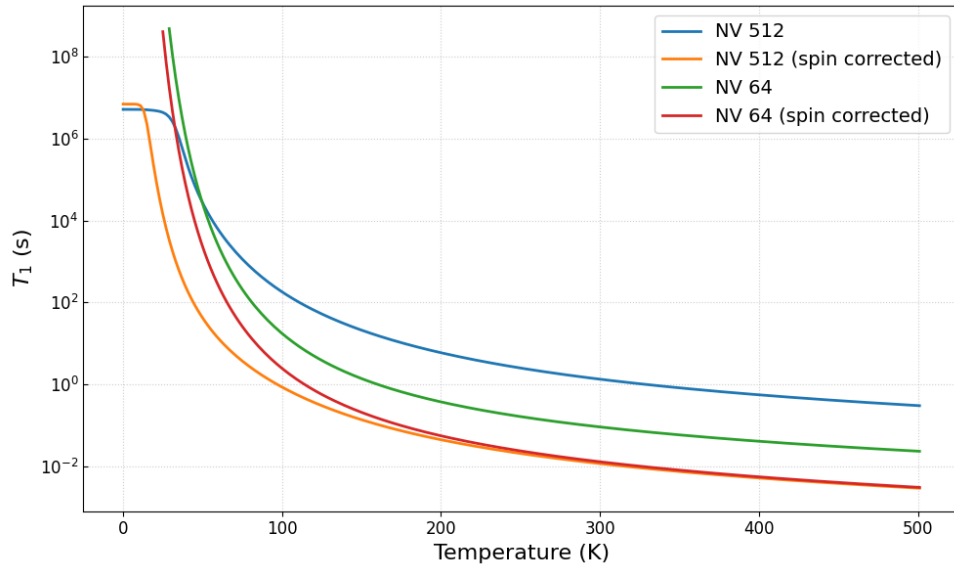


Figure 4.8: The T_1 relaxation time, for the $m_s = +1$ polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.

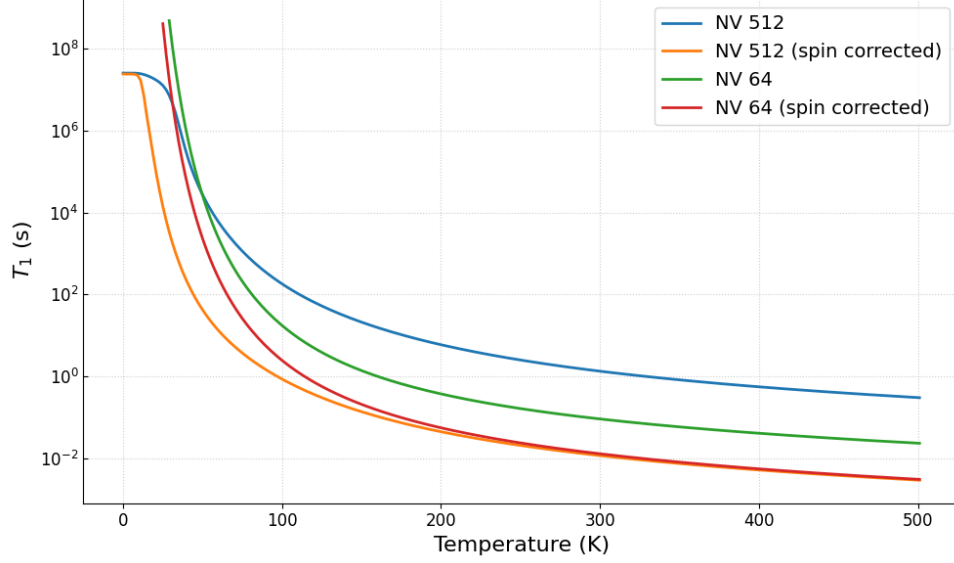


Figure 4.9: The T_1 relaxation time, for the $m_s = -1$, polarized state, plotted against temperature in a 64 and 512 atom NV center in diamond.

Table 4.1: T_1 relaxation time at room temperature (300 K) of a 64 and 512 atom NV center in diamond.

Defect	T_1 (ms)		
	$m_s = +1$	$m_s = 0$	$m_s = -1$
NV 512	1352	4118	1352
NV 512 (spin corrected)	11.72	8.297	11.72
NV 64	92.08	45.46	92.08
NV 64 (spin corrected)	13.00	7.534	13.00

4.2 CIV center in 4H-SiC

The similarities between the results from the 64 and 512 atom NV center motivated the choice of simulating a 128 atom supercell of CIV in SiC to reduce computation time as it still should provide a qualitative prediction of the relaxation time. The coupling coefficients, relaxation rates, and T_1 relaxation times for a CIV center in silicon carbide with supercell size of 128 atoms were calculated and are presented in the following sections. The perturbation applied to the atoms was $0.025 \text{ \AA} \sqrt{\text{a.m.u}}$ scaled with the eigenvectors of the phonon modes. The IPR was also calculated for the modes and is compared to the spin-phonon coupling coefficients.

Coupling coefficients and spectral function

The coupling strength is divided into two regions in the same structure as the NV center, with a much higher coupling strength for the acoustic phonons in first- and second-order interactions, as seen in Figures (4.10a) and (4.10c). The first-order coupling coefficients are smaller for the CIV center compared to the NV center but much higher in the second-order interactions.

The phonon energy is lower compared to the NV center by approximately 50 meV, as seen in Figure 4.10. The spin-phonon coupling for low energy phonons is therefore not negligible at the ZFS energy of $1.44 \text{ GHz} \approx 0.0060 \text{ meV}$.

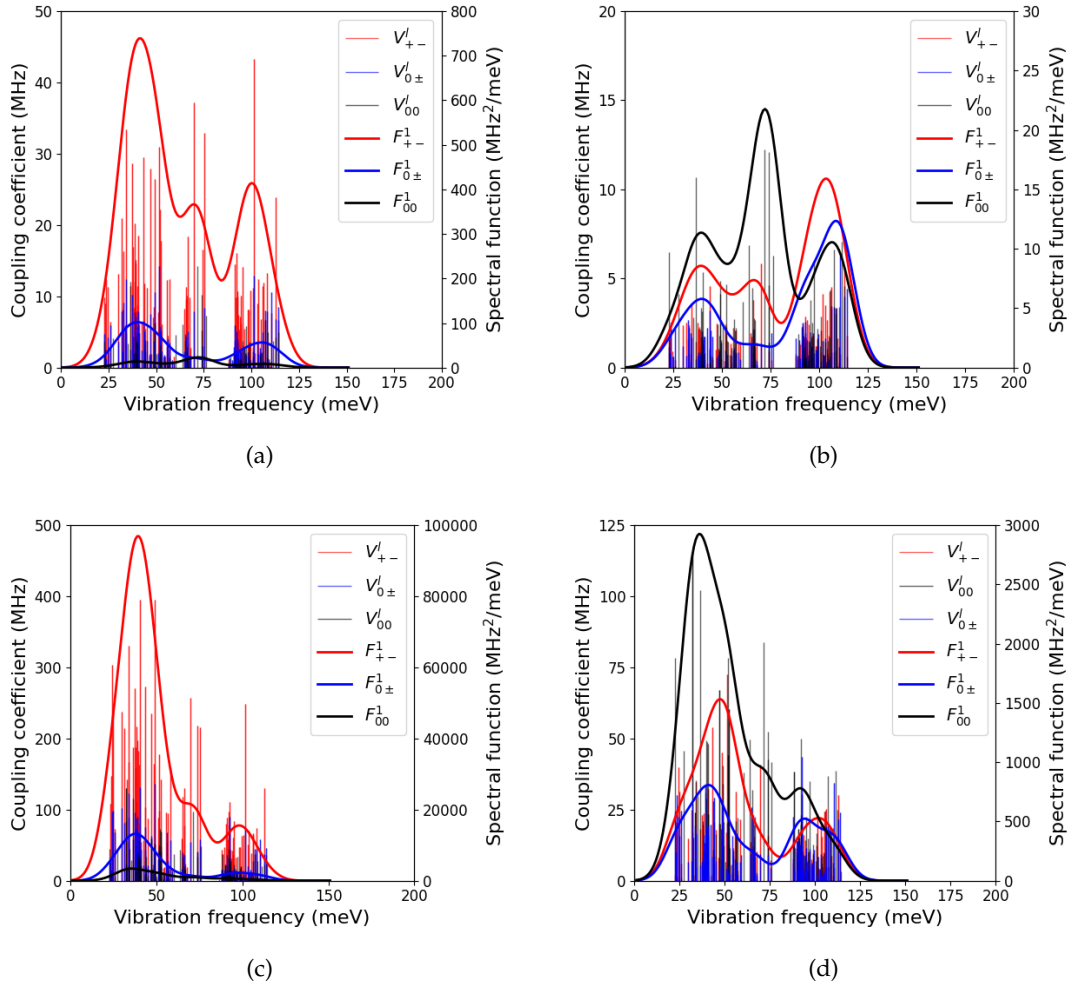


Figure 4.10: The coupling coefficients and spectral function for first and second order phonons for 128 atom supercell CIV center in 4H-SiC. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

Phonon mode localization vs coupling strength

The following figures show the coupling coefficients compared to IPR for the phonon modes in CIV. The same trends are noted for the CIV center, where no clear correlation can be seen between IPR and spin-phonon coupling strength. Instead a majority of the coupling is attributed to low IPR phonon modes as seen in Figure 4.11.

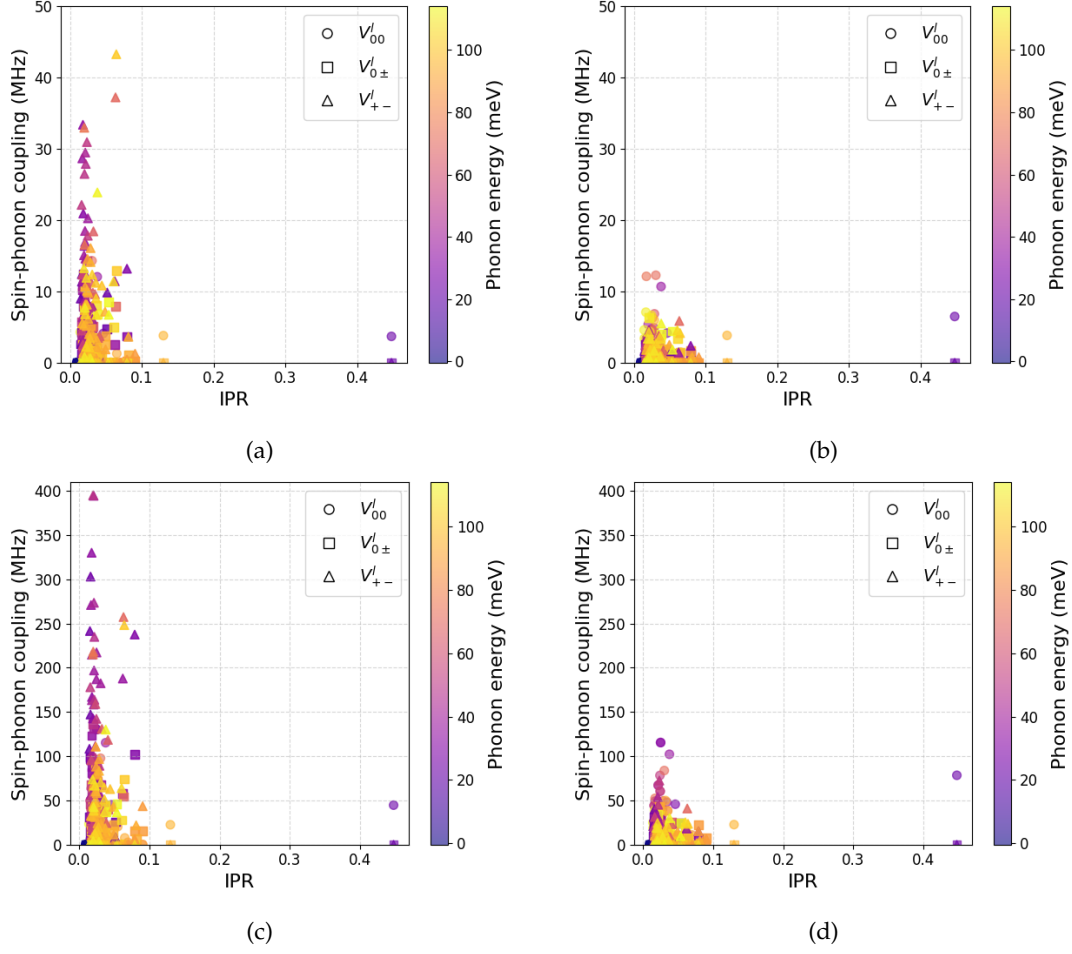


Figure 4.11: The first- and second-order spin-phonon coupling coefficients compared to the IPR in a 128 supercell CIV center. **(a)** Results for first-order coupling with spin contamination and **(b)** with spin-corrected ZFS. **(c)** Second-order spin-phonon coupling with spin contamination and **(d)** with spin-corrected ZFS.

Transition rates

In Figure 4.12, we see that the transition rate from the spin corrected ZFS is smaller than the spin contaminated rates, which is the opposite trend noted for the NV center. The first order and two phonon rates are constant up until $T \simeq 5$ K, after which they increase with the temperature. After 10 K, the first order rate has a linear form and the two phonon rate behaves as $T^{5/2}$. The magnitude of the transition rates tells us that both direct one phonon processes and two phonon Raman processes have major contributions to the T_1 time of the CIV center. In the low temperature region, the direct and double absorption/emission rates are in the order of ~ 1 mHz indicating a shorter T_1 time at low temperatures compared to the NV center.

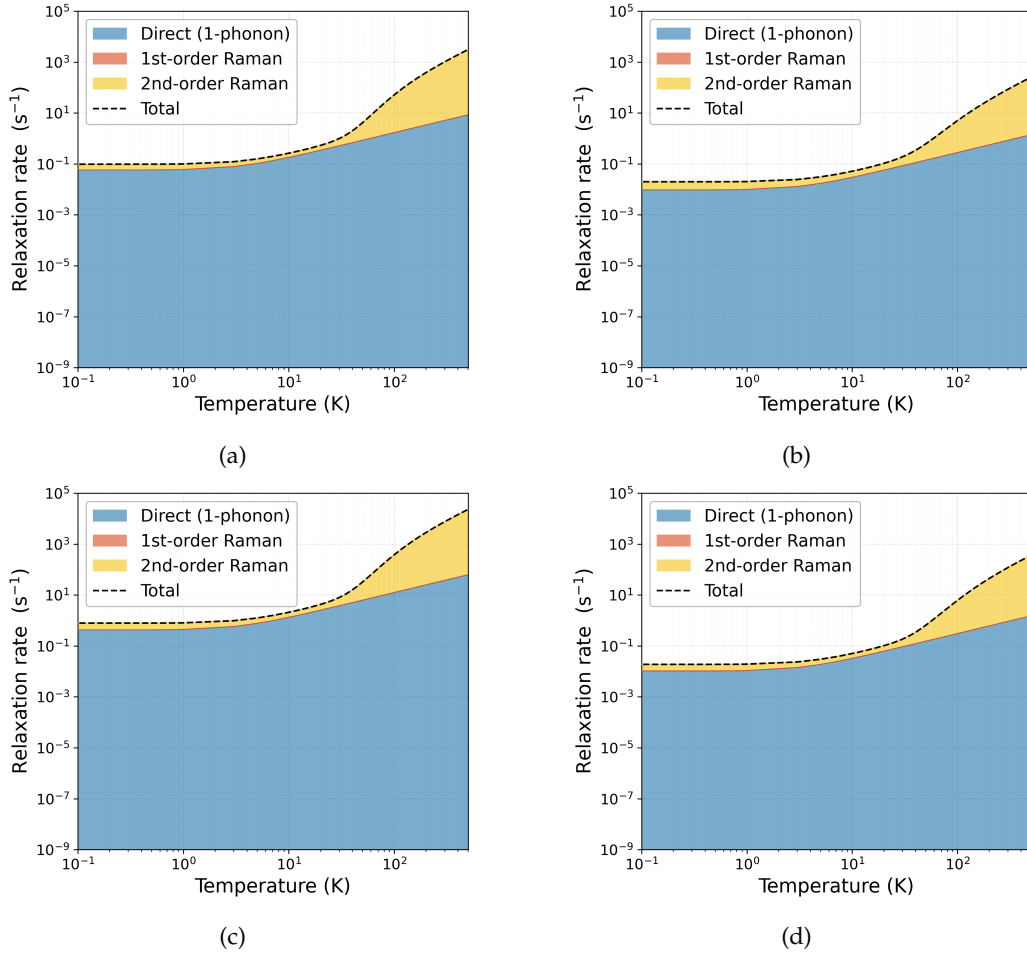


Figure 4.12: Transition rates for a 128 atom supercell size CIV center. **(a)** Single quantum transition rates ($\Gamma_{0\pm}$) with spin contaminated ZFS and **(b)** with spin corrected ZFS. **(c)** Double quantum transition rates (Γ_{+-}) with spin contaminated ZFS and **(d)** with spin corrected ZFS.

T_1 relaxation time

The relaxation time T_1 of the CIV center is a couple of seconds to a few hundred milliseconds in the low temperature range $T < 50$ K, see Figures (4.13) to (4.15). In the high temperature range $T > 300$ K, the defect has a relaxation time of a few hundred microseconds which is approximately 1000 times longer than the predicted time in Ref. [53]. The T_1 time in room temperature and at 0 K can be seen in Tables (4.2) and (4.3)

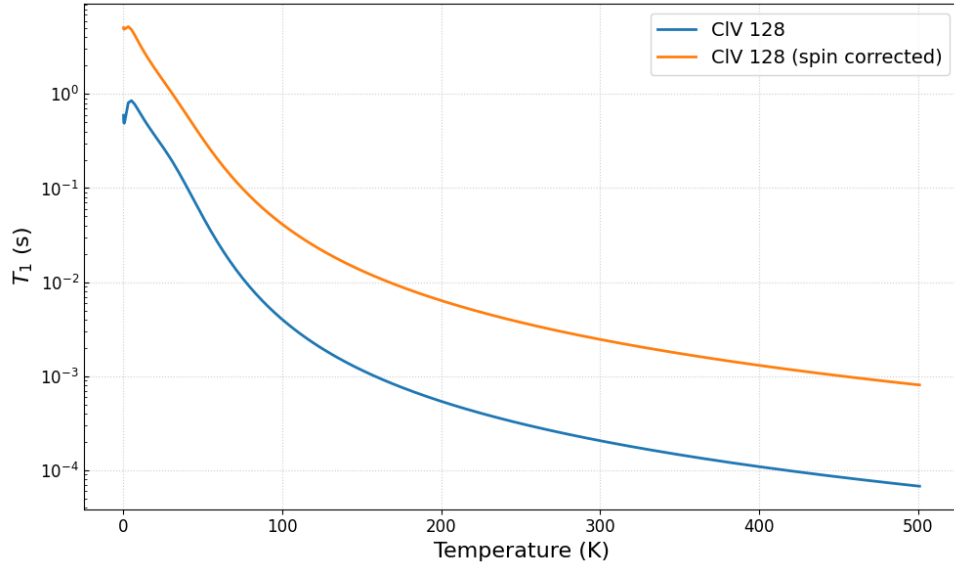


Figure 4.13: The T_1 relaxation time, for the $m_s = 0$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.

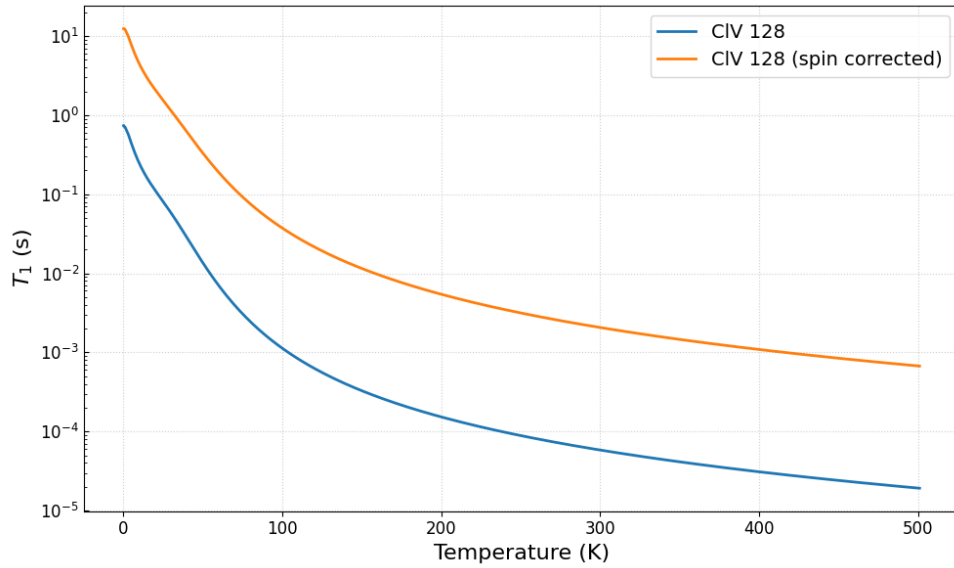


Figure 4.14: The T_1 relaxation time, for the $m_s = +1$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.

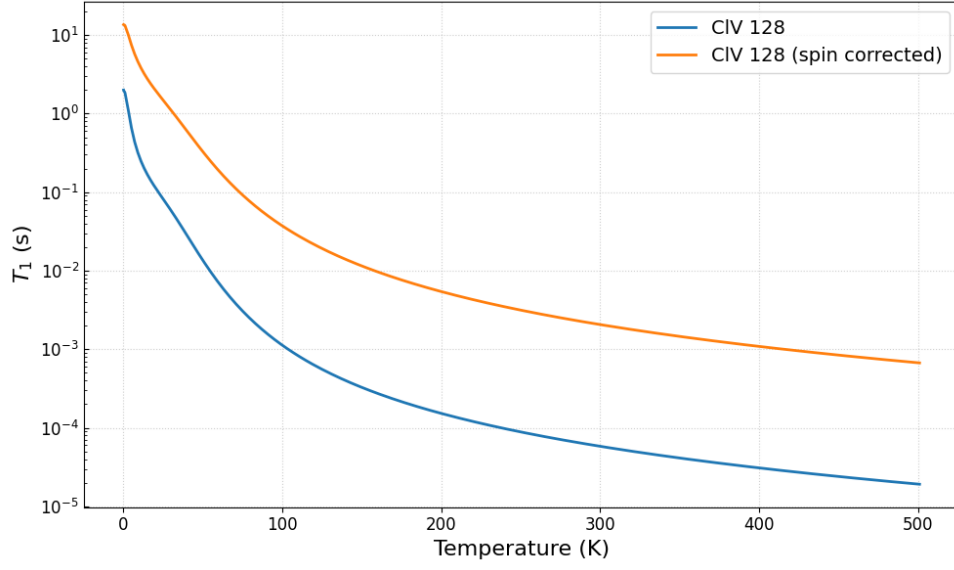


Figure 4.15: The T_1 relaxation time, for the $m_s = -1$ polarized state, plotted against temperature in a 128 atom CIV center in 4H-SiC.

Table 4.2: T_1 relaxation time at room temperature (300 K) of a 128 atom kk CIV center in 4H-SiC.

Defect	T_1 (μ s)		
	$m_s = +1$	$m_s = 0$	$m_s = -1$
CIV 128	58.69	207.6	58.69
CIV 128 (spin corrected)	2080	2478	2080

Table 4.3: T_1 relaxation time at 0 K of a 128 atom kk CIV center in 4H-SiC.

Defect	T_1 (s)		
	$m_s = +1$	$m_s = 0$	$m_s = -1$
CIV 128	0.7358	0.5941	1.997
CIV 128 (spin corrected)	12.40	5.063	13.57

5.1 Predicted stability of NV center in diamond

The NV center transition rates dominate different temperature regions. In the $T < 30\text{K}$ region, first order direct transitions dominate with a very low rate of $\sim 10\text{ }\mu\text{Hz}$. Second order phonon interactions $\Gamma_1^{(2)}$ clearly dominate the transition rate above 30 K and the ratio between Raman transitions from first order phonon interactions $\Gamma_2^{(1)}$ and second order phonon interactions is approximately $\Gamma_2^{(1)}/\Gamma_1^{(2)} \approx 10^{-6}$ which agrees with the analysis done by Cambria et al. [19].

Low energy phonons drive direct transitions in first-order processes and double absorption/emission. Because the limited supercell size used in this thesis restricts the density of low energy phonons, the true low-temperature relaxation rate is likely higher than predicted, i.e., the calculated rate is a lower bound. The temperature scaling, however, is not expected to be significantly affected, as the results in Figure 4.5 suggest that two-phonon Raman processes are dominant at high temperatures. The first order rates are much lower in the 64 atom supercell, see Figure 4.6, which is likely due to a sparser low energy phonon spectrum compared to the 512 atom supercell. The sparsity of low-energy phonons in the 64 atom supercell causes its spectral function to drop more sharply at low frequencies, suppressing the direct transition rate relative to the 512 atom cell.

Despite using the same methodology and phonon perturbation, the coupling coefficients of the NV center in the second order interactions are much larger than Cambria et al. [19]. Cambria et al. reports the largest coupling of 2 MHz at a phonon energy of 62.4 meV compared to the results in Figure 4.1c which show a strongest coupling of 58 MHz at 125.5 meV. The same discrepancy can be seen in the 64 atom supercell. However, it does not seem to impact the results of the second order transition rates of the 64 atom supercell seen in Figure 4.6 as the calculated transition rate is in the same order of magnitude as experimental data [14, 19, 50].

5.2 Predicted stability of CIV center in 4H-SiC

From Figure 4.12, we see that the 2nd-order Raman transitions contribute even at low temperatures, suggesting a non-zero spin-phonon coupling at the ZFS energy. Further verifying this theory, viewing the lower energy range of Figure 4.10, the two-phonon spectral function of CIV remains finite ($\sim 1\text{ Hz}$) at the ZFS energy, implying the existence of phonon pairs that match the ZFS energy even at zero temperature giving rise to double absorption/emission two-phonon processes, likely in the form of spontaneous emission. The direct and double absorption/emission processes provide a temperature-independent lower bound on the relaxation rate and upper bound on the T_1 time. The spectral function of the NV center is much smaller at the ZFS energy D indicating a weaker coupling of low energy phonons. The low energy phonon interactions suggests that the CIV center is a less suitable defect to use as a qubit compared to the NV center.

The high transition rate of the CIV center suggests a difficulty of initialising the qubit with optical pumping. If the rate $\Gamma_{0 \leftrightarrow \pm 1}$ is too high, spin relaxation will occur before a sufficient number of CIV centers are initialised.

The T_2 relaxation time has previously been estimated to $T_2 \approx 1.45\text{ ms}$ at $T = 0\text{ K}$ [21]. With the now computed temperature dependance of the T_1 time, it is noted that the previous estimation is reasonable as the relation between spin decoherence and spin-lattice relaxation

is $T_2 < 2T_1$ [8] and $T_1(T = 0) = 2.84$ s in Figure 4.15 for the $m_s = -1$ polarized state, which is the suspected most stable qubit state [21].

5.3 Spin contamination and the connection to symmetry

The coupling coefficients of the spin-decontaminated calculations are, in the first order interactions, clearly smaller and have a different shape than the coefficients of the spin contaminated results. The spin-phonon coupling coefficients $V_{0\pm}^{II}$ and V_{+-}^{II} are impacted the most, as seen in Figures (4.1), (4.2) and (4.10), indicating that the spin-phonon coupling of phonon modes transforming as E get a large contribution from valence band orbitals. One possible explanation is that E -modes couple more strongly to the valence-band orbitals, which are omitted in the spin-decontaminated DFT treatment. The presence of many delocalized E and A_1 modes in Figures (4.3), (4.4) and (4.11) suggests, however, that V_{00}^{II} would also be reduced if the effect originated from bulk-orbital contributions alone. When the spin-phonon coupling of $V_{0\pm}^{II}$ and V_{+-}^{II} are reduced, the transition rates are also affected.

After decontamination, the ordering of the single- and double-quantum transitions reverses, see Figures (4.6b) and (4.6d). Experimental data shows that the rates of double quantum transitions are higher than single quantum transitions [14, 19, 50] for the NV center. This reversed ordering of the rates is also observed for the CIV center, but the spin contaminated T_1 time is lower, not higher, than the spin corrected T_1 time, see Figure 4.15, which is the opposite of what is observed for the NV center, see Figure 4.7. The connection between symmetry breaking modes and bulk spins warrants further investigation through a more detailed group theoretical analysis. It would also be interesting to determine if other defects exhibiting C_{3v} symmetry display similar effects.

A reason for the connection between symmetry breaking modes and spin contamination could be because symmetry breaking modes lower the symmetry of potential A_1 electron orbitals in the valence band, which contribute to the ZFS when there is spin contamination. This valence band contribution to the ZFS is not present in the spin corrected method and therefore the symmetry breaking modes impact on the bulk orbitals are not as important.

5.4 Analysis of the IPR

No systematic correlation between IPR and spin-phonon coupling emerges from Figures (4.3), (4.4) and (4.11). A few modes exhibit both high IPR and high spin-phonon coupling, but as noted by Cambria et al. this may be coincidental. Rather, it is regions with many delocalized modes with moderate coupling that likely dominate the relaxation.

5.5 Method

As mentioned in the previous section, the spectral function of the 64 atom supercell NV center drops more sharply than the 512 atom cell. This drop could be compensated by utilizing a higher standard deviation σ for the gaussian smearing in order to get a higher contribution from the low energy modes in the direct transitions but the choice of standard deviation must be physically motivated. In this thesis, the choice of $\sigma = 7.5$ meV is based upon Cambria et al. and not derived methodically. This choice significantly affects the spectral function and therefore the relative magnitudes of different transition rates.

The empirical spin decontamination method used here has been validated only on a limited set of equilibrium structures and has not been tested for displaced geometries, introducing considerable uncertainty. One should compare the full spin decontamination method suggested by Biktagirov et al. [25] by calculating the ZFS of the $S' = S - 1$ states as it would present more details on whether the connection between bulk orbitals and E -mode phonons are due to the specific correction method used or if there exists a physical

connection. Applying the full spin decontamination method would however require more computational time and thus would be outside the scope of this thesis due to limited computational resources. One could also investigate other methods of spin decontamination, e.g., the one suggested in ref. [26] to see if the effect still arises.

The computational framework builds heavily on the method of Cambria et al. [19], which is used by others as well, with some variations [36]. The Taylor expansion to second order of the spin hamiltonian neglects higher order terms. Expanding further is computationally heavy, if not infeasible, would require additional group theoretical analysis and is therefore left out of this thesis but it is important to note that higher order phonon interactions could be relevant to capture anharmonic effects of the phonon modes. Direct experimental values of spin-phonon coupling coefficients are not available and therefore the validation of results relies on the accuracy of derived transition rates from temperature dependent T_1 measurements. The results are therefore most robust when interpreted in terms of relative trends and comparisons in orders of magnitude, rather than as precise predictions of individual coupling coefficients.

5.6 The work in a wider context

This thesis shows that the CIV center in 4H-SiC has a lower T_1 time compared to the NV center in diamond making it difficult to use in large gate arrays and quantum algorithms as the qubit state would be lost during the computation. However, the CIV is still a promising point defect for quantum applications, mainly for its telecom band zero-phonon line, making it integrable in modern communication technology.

5.7 Outlook

Because of the computational cost of DFT, mainly the computation of phonons, the simulations in this thesis are limited to a supercell size of 512 atoms for the NV center and 128 atoms for the CIV center. Machine learned interatomic potentials (MLIPs) would enable an analysis of larger supercells which would introduce more low-energy phonons driving direct spin transitions. A future study could compare the resulting relaxation rates using phonons calculated with DFT and a MLIP and see if the results are qualitatively the same. If that is the case, it will increase the speed of high-throughput screening of defect materials suitable for hosting qubits.

The unexpected sensitivity of spin-contaminated coupling coefficients to E -symmetry modes motivates a systematic group-theoretical investigation across other C_{3v} defects, such as the divacancy in SiC, to determine whether this is a general feature.

Extending the strain engineering approach of Ahn et al. [54] to the CIV center is a further logical step as it would test whether strain could mitigate the strong spin-phonon coupling of the CIV center that limits the T_1 time.

CONCLUSION

Understanding the spin-relaxation mechanisms in point defects is crucial for evaluating the viability of defect qubits. This thesis investigates the relaxation rate and T_1 relaxation time of the NV center in diamond and CIV center in 4H-SiC. The thesis also investigates how spin contamination in DFT methods affect the transition rates driven by spin-phonon coupling, and whether local phonon modes dominate these transitions. The localization of phonon modes is calculated using IPR.

First, the IPR shows no clear correlation with the spin-phonon coupling coefficients of the corresponding modes and the results disprove the hypothesis that local phonon modes drive spin transitions. Instead, it is found that a high density of non-local bulk modes dominate the transition rate and contribution to the T_1 time.

In unrestricted DFT calculations, the spatial relaxation of the spin channels leads to spin contamination - the wave function is no longer an eigenfunction to the S^2 operator. Spin contamination causes the ZFS to be influenced not only by the unpaired defect spins but also by spins in the valence band. Spin contaminated states appears to have a strong coupling to phonon modes that transform according to the E irrep of the C_{3v} point group.

The model predicts a relaxation time of $7.534 \text{ ms} \leq T_1 \leq 45.46 \text{ ms}$ for the NV center in diamond at room temperature using the 64 atom supercell, in agreement with previous experimental and theoretical values. It also predicts the room temperature T_1 relaxation time of CIV in 4H-SiC to be $58.69 \mu\text{s} \leq T_1 \leq 2.080 \text{ ms}$, a property still requiring experimental confirmation. The relaxation time of CIV in 4H-SiC is found to be significantly lower than the relaxation time of the NV center in diamond, making it less suitable to use as a qubit, at least in the spin contaminated formalism. Two main factors explain this difference; first, the phonons of the CIV center are lower in energy, making the defect more sensitive to temperature; second, the CIV center exhibits a substantially larger spin-phonon coupling. Our results show that the second order spin-phonon coupling is approximately ten times greater in the CIV center than in the NV center. However, further studies are needed to understand if these effects are mainly related to the CIV defect or all defects in 4H-SiC. Furthermore, one could address the high temperature sensitivity by applying external strain to the crystal to shift the phonon band structure.

From the results presented in this thesis, it is apparent that further research is required on the phenomenon of spin contamination and the spin-phonon interaction. Applying restricted open-shell DFT methods could decrease spin contamination and provide deeper insight of this effect to spin-phonon coupling. This thesis highlights the importance of accurate spin-phonon coupling calculations, reveals the adverse effects of spin contamination on defect spin dynamics, and provides a rigorous framework for testing the stability of point-defect qubits in wide band gap semiconductor materials.

BIBLIOGRAPHY

- [1] Charles H. Bennett and Gilles Brassard. “Quantum cryptography: Public key distribution and coin tossing”. en. In: *Theoretical Computer Science* 560 (Dec. 2014), pp. 7–11. ISSN: 03043975. DOI: 10 . 1016 / j . tcs . 2014 . 05 . 025. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0304397514004241> (visited on 05/22/2026).
- [2] Marta Gilaberte Basset, Frank Setzpfandt, Fabian Steinlechner, Erik Beckert, Thomas Pertsch, and Markus Gräfe. “Perspectives for Applications of Quantum Imaging”. en. In: *Laser & Photonics Reviews* 13.10 (Oct. 2019), p. 1900097. ISSN: 1863-8880, 1863-8899. DOI: 10.1002/lpor.201900097. URL: <https://onlinelibrary.wiley.com/doi/10.1002/lpor.201900097> (visited on 05/22/2026).
- [3] Romana Schirhagl, Kevin Chang, Michael Loretz, and Christian L. Degen. “Nitrogen-Vacancy Centers in Diamond: Nanoscale Sensors for Physics and Biology”. en. In: *Annual Review of Physical Chemistry* 65.1 (Apr. 2014), pp. 83–105. ISSN: 0066-426X, 1545-1593. DOI: 10.1146/annurev-physchem-040513-103659. URL: <https://www.annualreviews.org/doi/10.1146/annurev-physchem-040513-103659> (visited on 01/28/2026).
- [4] Bruce M. Boghosian and Washington Taylor. “Simulating quantum mechanics on a quantum computer”. en. In: *Physica D: Nonlinear Phenomena* 120.1-2 (Sept. 1998), pp. 30–42. ISSN: 01672789. DOI: 10 . 1016 / S0167 - 2789 (98) 00042 - 6. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0167278998000426> (visited on 05/04/2026).
- [5] Hannes Hübel, Michael R. Vanner, Thomas Lederer, Bibiane Blauensteiner, Thomas Lorünser, Andreas Poppe, and Anton Zeilinger. “High-fidelity transmission of polarization encoded qubits from an entangled source over 100 km of fiber”. en. In: *Optics Express* 15.12 (2007), p. 7853. ISSN: 1094-4087. DOI: 10.1364/OE.15.007853. URL: <https://opg.optica.org/oe/abstract.cfm?uri=oe-15-12-7853> (visited on 06/01/2026).
- [6] Lucio Robledo, Lilian Childress, Hannes Bernien, Bas Hensen, Paul F. A. Alkemade, and Ronald Hanson. “High-fidelity projective read-out of a solid-state spin quantum register”. en. In: *Nature* 477.7366 (Sept. 2011), pp. 574–578. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/nature10401. URL: <https://www.nature.com/articles/nature10401> (visited on 02/11/2026).
- [7] Nathalie P. De Leon, Kohei M. Itoh, Dohun Kim, Karan K. Mehta, Tracy E. Northup, Hanhee Paik, B. S. Palmer, N. Samarth, Sorawis Sangtawesin, and D. W. Steuerman. “Materials challenges and opportunities for quantum computing hardware”. en. In: *Science* 372.6539 (Apr. 2021), eabb2823. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.abb2823. URL: <https://www.science.org/doi/10.1126/science.abb2823> (visited on 01/30/2026).
- [8] Daniel D. Traficante. “Relaxation. Can T_2 , be longer than T_1 ?” en. In: *Concepts in Magnetic Resonance* 3.3 (July 1991), pp. 171–177. ISSN: 1043-7347, 1099-0534. DOI: 10 . 1002 / cmr . 1820030305. URL: <https://onlinelibrary.wiley.com/doi/10.1002/cmr.1820030305> (visited on 05/27/2026).

- [9] N. Bar-Gill, L.M. Pham, A. Jarmola, D. Budker, and R.L. Walsworth. "Solid-state electronic spin coherence time approaching one second". en. In: *Nature Communications* 4.1 (Apr. 2013), p. 1743. ISSN: 2041-1723. DOI: 10.1038/ncomms2771. URL: <https://www.nature.com/articles/ncomms2771> (visited on 05/27/2026).
- [10] A. M. Tyryshkin, S. A. Lyon, A. V. Astashkin, and A. M. Raitsimring. "Electron spin relaxation times of phosphorus donors in silicon". en. In: *Physical Review B* 68.19 (Nov. 2003), p. 193207. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.68.193207. URL: <https://link.aps.org/doi/10.1103/PhysRevB.68.193207> (visited on 05/27/2026).
- [11] F. Borjans, D.M. Zajac, T.M. Hazard, and J.R. Petta. "Single-Spin Relaxation in a Synthetic Spin-Orbit Field". en. In: *Physical Review Applied* 11.4 (Apr. 2019), p. 044063. ISSN: 2331-7019. DOI: 10.1103/PhysRevApplied.11.044063. URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.11.044063> (visited on 05/22/2026).
- [12] Kenneth R. Brown, John Chiaverini, Jeremy Sage, and Hartmut Häffner. *Materials Challenges for Trapped-Ion Quantum Computers*. 2020. DOI: 10.48550/ARXIV.2009.00568. URL: <https://arxiv.org/abs/2009.00568> (visited on 06/01/2026).
- [13] Marcus W. Doherty, Neil B. Manson, Paul Delaney, Fedor Jelezko, Jörg Wrachtrup, and Lloyd C.L. Hollenberg. "The nitrogen-vacancy colour centre in diamond". en. In: *Physics Reports* 528.1 (July 2013), pp. 1–45. ISSN: 03701573. DOI: 10.1016/j.physrep.2013.02.001. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0370157313000562> (visited on 02/16/2026).
- [14] M. C. Cambria, A. Gardill, Y. Li, A. Norambuena, J. R. Maze, and S. Kolkowitz. "State-dependent phonon-limited spin relaxation of nitrogen-vacancy centers". en. In: *Physical Review Research* 3.1 (Feb. 2021), p. 013123. ISSN: 2643-1564. DOI: 10.1103/PhysRevResearch.3.013123. URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.3.013123> (visited on 01/14/2026).
- [15] Joel Davidsson, Viktor Ivády, Rickard Armiento, and Igor A. Abrikosov. "ADAQ: Automatic workflows for magneto-optical properties of point defects in semiconductors". en. In: *Computer Physics Communications* 269 (Dec. 2021), p. 108091. ISSN: 00104655. DOI: 10.1016/j.cpc.2021.108091. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0010465521002034> (visited on 02/16/2026).
- [16] A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. Von Borczyskowski. "Scanning Confocal Optical Microscopy and Magnetic Resonance on Single Defect Centers". en. In: *Science* 276.5321 (June 1997), pp. 2012–2014. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.276.5321.2012. URL: <https://www.science.org/doi/10.1126/science.276.5321.2012> (visited on 02/16/2026).
- [17] David D. Awschalom, Ronald Hanson, Jörg Wrachtrup, and Brian B. Zhou. "Quantum technologies with optically interfaced solid-state spins". en. In: *Nature Photonics* 12.9 (Sept. 2018), pp. 516–527. ISSN: 1749-4885, 1749-4893. DOI: 10.1038/s41566-018-0232-2. URL: <https://www.nature.com/articles/s41566-018-0232-2> (visited on 02/10/2026).

- [18] Alessandro Lunghi and Stefano Sanvito. "How do phonons relax molecular spins?" en. In: *Science Advances* 5.9 (Sept. 2019), eaax7163. ISSN: 2375-2548. DOI: 10.1126/sciadv.aax7163. URL: <https://www.science.org/doi/10.1126/sciadv.aax7163> (visited on 01/12/2026).
- [19] M. C. Cambria, A. Norambuena, H. T. Dinani, G. Thiering, A. Gardill, I. Kemeny, Y. Li, V. Lordi, Á. Gali, J. R. Maze, and S. Kolkowitz. "Temperature-Dependent Spin-Lattice Relaxation of the Nitrogen-Vacancy Spin Triplet in Diamond". en. In: *Physical Review Letters* 130.25 (June 2023), p. 256903. ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.130.256903. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.130.256903> (visited on 01/13/2026).
- [20] J. Gugler, T. Astner, A. Angerer, J. Schmiedmayer, J. Majer, and P. Mohn. "Ab initio calculation of the spin lattice relaxation time T_1 for nitrogen-vacancy centers in diamond". en. In: *Physical Review B* 98.21 (Dec. 2018), p. 214442. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.98.214442. URL: <https://link.aps.org/doi/10.1103/PhysRevB.98.214442> (visited on 01/13/2026).
- [21] Oscar Balancea-Lindvall, Joel Davidsson, Rickard Armiento, and Igor A. Abrikosov. "Chlorine vacancy in 4 H - SiC : An NV-like defect with telecom-wavelength emission". en. In: *Physical Review B* 108.22 (Dec. 2023), p. 224106. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.108.224106. URL: <https://link.aps.org/doi/10.1103/PhysRevB.108.224106> (visited on 01/30/2026).
- [22] Andrey N. Anisimov, Ashin V. Mathews, Kalliopi Mavridou, Ulrich Kentsch, Manfred Helm, and Georgy V. Astakhov. "Engineering chlorine-based emitters in silicon carbide for telecom-band quantum technologies". en. In: *Optics Express* 33.26 (Dec. 2025), p. 54285. ISSN: 1094-4087. DOI: 10.1364/OE.581705. URL: <https://opg.optica.org/abstract.cfm?URI=oe-33-26-54285> (visited on 01/30/2026).
- [23] Patrick Jost and Christoph Van Wüllen. "Why spin contamination is a major problem in the calculation of spin-spin coupling in triplet biradicals". en. In: *Physical Chemistry Chemical Physics* 15.39 (2013), p. 16426. ISSN: 1463-9076, 1463-9084. DOI: 10.1039/c3cp52568f. URL: <https://xlink.rsc.org/?DOI=c3cp52568f> (visited on 05/12/2026).
- [24] Jon Baker, Andrew Scheiner, and Jan Andzelm. "Spin contamination in density functional theory". en. In: *Chemical Physics Letters* 216.3-6 (Dec. 1993), pp. 380-388. ISSN: 00092614. DOI: 10.1016/0009-2614(93)90113-F. URL: <https://linkinghub.elsevier.com/retrieve/pii/000926149390113F> (visited on 05/12/2026).
- [25] Timur Biktagirov, Wolf Gero Schmidt, and Uwe Gerstmann. "Spin decontamination for magnetic dipolar coupling calculations: Application to high-spin molecules and solid-state spin qubits". en. In: *Physical Review Research* 2.2 (Apr. 2020), p. 022024. ISSN: 2643-1564. DOI: 10.1103/PhysRevResearch.2.022024. URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.2.022024> (visited on 04/17/2026).
- [26] Nicolas Ferré, Nathalie Guihéry, and Jean-Paul Malrieu. "Spin decontamination of broken-symmetry density functional theory calculations: deeper insight and new formulations". en. In: *Physical Chemistry Chemical Physics* 17.22 (2015), pp. 14375-14382. ISSN: 1463-9076, 1463-9084. DOI: 10.1039/C4CP05531D. URL: <https://xlink.rsc.org/?DOI=C4CP05531D> (visited on 05/12/2026).

- [27] Adam Gali. *Ab initio theory of nitrogen-vacancy center in diamond*. 2019. DOI: 10.48550/ARXIV.1906.00047. URL: <https://arxiv.org/abs/1906.00047> (visited on 05/22/2026).
- [28] Zsolt Benedek, Oscar Bulancea-Lindvall, Joel Davidsson, Viktor Ivády, and Igor A. Abrikosov. "Spin-dependent optical activity of the telecom range CIV center in SiC: a wavefunction theory analysis". en. In: *npj Quantum Materials* (May 2026). ISSN: 2397-4648. DOI: 10.1038/s41535-026-00888-2. URL: <https://www.nature.com/articles/s41535-026-00888-2> (visited on 06/25/2026).
- [29] Zsolt Benedek, Oscar Bulancea-Lindvall, Joel Davidsson, Viktor Ivády, and Igor Abrikosov. *Demonstration of the ODMR activity of the telecom range CIV center in SiC: a wavefunction theory analysis*. arXiv:2511.21965. Nov. 2025. DOI: 10.48550/arXiv.2511.21965. URL: <http://arxiv.org/abs/2511.21965> (visited on 01/30/2026).
- [30] Charles Kittel. *Introduction to solid state physics*. eng. 8. ed., [repr.] Hoboken, NJ: Wiley. ISBN: 9780471415268.
- [31] John E. Harriman and Ernest M. Loebl. *Theoretical Foundations of Electron Spin Resonance: Physical Chemistry: A Series of Monographs*. eng. Burlington: Elsevier Science, 1978. ISBN: 9780123263506.
- [32] Adam Gali, Maria Fyta, and Efthimios Kaxiras. "Ab initio supercell calculations on nitrogen-vacancy center in diamond: Electronic structure and hyperfine tensors". en. In: *Physical Review B* 77.15 (Apr. 2008), p. 155206. ISSN: 1098-0121, 1550-235X. DOI: 10.1103/PhysRevB.77.155206. URL: <https://link.aps.org/doi/10.1103/PhysRevB.77.155206> (visited on 06/02/2026).
- [33] S. Felton, A. M. Edmonds, M. E. Newton, P. M. Martineau, D. Fisher, D. J. Twitchen, and J. M. Baker. "Hyperfine interaction in the ground state of the negatively charged nitrogen vacancy center in diamond". en. In: *Physical Review B* 79.7 (Feb. 2009), p. 075203. ISSN: 1098-0121, 1550-235X. DOI: 10.1103/PhysRevB.79.075203. URL: <https://link.aps.org/doi/10.1103/PhysRevB.79.075203> (visited on 02/11/2026).
- [34] Viktor Ivády, Igor A. Abrikosov, and Adam Gali. "First principles calculation of spin-related quantities for point defect qubit research". en. In: *npj Computational Materials* 4.1 (Dec. 2018), p. 76. ISSN: 2057-3960. DOI: 10.1038/s41524-018-0132-5. URL: <https://www.nature.com/articles/s41524-018-0132-5> (visited on 02/03/2026).
- [35] Lorenzo A. Mariano, Vu Ha Anh Nguyen, Jonatan B. Petersen, Magnus Björnsson, Jesper Bendix, Gareth R. Eaton, Sandra S. Eaton, and Alessandro Lunghi. "The role of electronic excited states in the spin-lattice relaxation of spin-1/2 molecules". en. In: *Science Advances* 11.7 (Feb. 2025), eadr0168. ISSN: 2375-2548. DOI: 10.1126/sciadv.adr0168. URL: <https://www.science.org/doi/10.1126/sciadv.adr0168> (visited on 01/11/2026).
- [36] A. Norambuena, E. Muñoz, H. T. Dinani, A. Jarmola, P. Maletinsky, D. Budker, and J. R. Maze. "Spin-lattice relaxation of individual solid-state spins". en. In: *Physical Review B* 97.9 (Mar. 2018), p. 094304. ISSN: 2469-9950, 2469-9969. DOI: 10.1103/PhysRevB.97.094304. URL: <https://link.aps.org/doi/10.1103/PhysRevB.97.094304> (visited on 03/11/2026).

- [37] Robert G. Parr and Yang Weitao. *Density-Functional Theory of Atoms and Molecules*. en. Oxford University Press, Jan. 1995. ISBN: 9780195092769. DOI: 10.1093/oso/9780195092769.001.0001. URL: <https://academic.oup.com/book/41995> (visited on 04/27/2026).
- [38] P. Hohenberg and W. Kohn. "Inhomogeneous Electron Gas". en. In: *Physical Review* 136.3B (Nov. 1964), B864–B871. ISSN: 0031-899X. DOI: 10.1103/PhysRev.136.B864. URL: <https://link.aps.org/doi/10.1103/PhysRev.136.B864> (visited on 04/27/2026).
- [39] W. Kohn and L. J. Sham. "Self-Consistent Equations Including Exchange and Correlation Effects". en. In: *Physical Review* 140.4A (Nov. 1965), A1133–A1138. ISSN: 0031-899X. DOI: 10.1103/PhysRev.140.A1133. URL: <https://link.aps.org/doi/10.1103/PhysRev.140.A1133> (visited on 04/27/2026).
- [40] John P. Perdew, Matthias Ernzerhof, and Kieron Burke. "Rationale for mixing exact exchange with density functional approximations". en. In: *The Journal of Chemical Physics* 105.22 (Dec. 1996), pp. 9982–9985. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.472933. URL: <https://pubs.aip.org/jcp/article/105/22/9982/478038/Rationale-for-mixing-exact-exchange-with-density> (visited on 05/27/2026).
- [41] Jochen Heyd, Gustavo E. Scuseria, and Matthias Ernzerhof. "Hybrid functionals based on a screened Coulomb potential". en. In: *The Journal of Chemical Physics* 118.18 (May 2003), pp. 8207–8215. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.1564060. URL: <https://pubs.aip.org/jcp/article/118/18/8207/460359/Hybrid-functionals-based-on-a-screened-Coulomb> (visited on 05/27/2026).
- [42] Aron J. Cohen, David J. Tozer, and Nicholas C. Handy. "Evaluation of $\langle S^2 \rangle$ in density functional theory". en. In: *The Journal of Chemical Physics* 126.21 (June 2007), p. 214104. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.2737773. URL: <https://pubs.aip.org/jcp/article/126/21/214104/187028/Evaluation-of-S-2-in-density-functional-theory> (visited on 05/14/2026).
- [43] G. Kresse and J. Hafner. "*Ab initio* molecular dynamics for liquid metals". en. In: *Physical Review B* 47.1 (Jan. 1993), pp. 558–561. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.47.558. URL: <https://link.aps.org/doi/10.1103/PhysRevB.47.558> (visited on 02/06/2026).
- [44] G. Kresse and J. Hafner. "*Ab initio* molecular-dynamics simulation of the liquid-metal-amorphous-semiconductor transition in germanium". en. In: *Physical Review B* 49.20 (May 1994), pp. 14251–14269. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.49.14251. URL: <https://link.aps.org/doi/10.1103/PhysRevB.49.14251> (visited on 02/06/2026).
- [45] G. Kresse and J. Furthmüller. "Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set". en. In: *Computational Materials Science* 6.1 (July 1996), pp. 15–50. ISSN: 09270256. DOI: 10.1016/0927-0256(96)00008-0. URL: <https://linkinghub.elsevier.com/retrieve/pii/0927025696000080> (visited on 02/06/2026).

- [46] G. Kresse and J. Furthmüller. “Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set”. en. In: *Physical Review B* 54.16 (Oct. 1996), pp. 11169–11186. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.54.11169. URL: <https://link.aps.org/doi/10.1103/PhysRevB.54.11169> (visited on 02/06/2026).
- [47] G Kresse and J Hafner. “Norm-conserving and ultrasoft pseudopotentials for first-row and transition elements”. In: *Journal of Physics: Condensed Matter* 6.40 (Oct. 1994), pp. 8245–8257. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/0953-8984/6/40/015. URL: <https://iopscience.iop.org/article/10.1088/0953-8984/6/40/015> (visited on 02/06/2026).
- [48] G. Kresse and D. Joubert. “From ultrasoft pseudopotentials to the projector augmented-wave method”. en. In: *Physical Review B* 59.3 (Jan. 1999), pp. 1758–1775. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.59.1758. URL: <https://link.aps.org/doi/10.1103/PhysRevB.59.1758> (visited on 02/06/2026).
- [49] Shyue Ping Ong, William Davidson Richards, Anubhav Jain, Geoffroy Hautier, Michael Kocher, Shreyas Cholia, Dan Gunter, Vincent L. Chevrier, Kristin A. Persson, and Gerbrand Ceder. “Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis”. en. In: *Computational Materials Science* 68 (Feb. 2013), pp. 314–319. ISSN: 09270256. DOI: 10.1016/j.commatsci.2012.10.028. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0927025612006295> (visited on 04/20/2026).
- [50] A. Gardill, M.C. Cambria, and S. Kolkowitz. “Fast Relaxation on Qutrit Transitions of Nitrogen-Vacancy Centers in Nanodiamonds”. en. In: *Physical Review Applied* 13.3 (Mar. 2020), p. 034010. ISSN: 2331-7019. DOI: 10.1103/PhysRevApplied.13.034010. URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.13.034010> (visited on 05/21/2026).
- [51] T. Astner, J. Gugler, A. Angerer, S. Wald, S. Putz, N. J. Mauser, M. Trupke, H. Sumiya, S. Onoda, J. Isoya, J. Schmiedmayer, P. Mohn, and J. Majer. “Solid-state electron spin lifetime limited by phononic vacuum modes”. en. In: *Nature Materials* 17.4 (Apr. 2018), pp. 313–317. ISSN: 1476-1122, 1476-4660. DOI: 10.1038/s41563-017-0008-y. URL: <https://www.nature.com/articles/s41563-017-0008-y> (visited on 02/10/2026).
- [52] D. A. Redman, S. Brown, R. H. Sands, and S. C. Rand. “Spin dynamics and electronic states of N- V centers in diamond by EPR and four-wave-mixing spectroscopy”. en. In: *Physical Review Letters* 67.24 (Dec. 1991), pp. 3420–3423. ISSN: 0031-9007. DOI: 10.1103/PhysRevLett.67.3420. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.67.3420> (visited on 05/26/2026).
- [53] A. N. Anisimov, K. Mavridou, A. V. Mathews, M. Helm, and G. V. Astakhov. *Telecom-band quantum memory with chlorine defects in silicon carbide*. 2026. DOI: 10.48550/ARXIV.2605.03717. URL: <https://arxiv.org/abs/2605.03717> (visited on 05/08/2026).
- [54] Jonghoon Ahn, Christina Wicker, Nolan Bitner, Michael T. Solomon, Benedikt Tissot, Guido Burkard, Alan M. Dibos, Jiefei Zhang, F. Joseph Heremans, and David D. Awschalom. “Extended spin relaxation times of optically addressed vanadium defects in silicon carbide at telecommunication frequencies”. en. In: *Physical Review Applied* 22.4 (Oct. 2024), p. 044078. ISSN: 2331-7019. DOI: 10.1103/PhysRevApplied.22.044078. URL: <https://link.aps.org/doi/10.1103/PhysRevApplied.22.044078> (visited on 05/12/2026).

MATHEMATICAL DERIVATIONS

A.1 Zero Field splitting

The ZFS tensor can be simplified to a description with only two variables D and E . The following derivation shows how to reach this description. The ZFS Hamiltonian can be written as

$$\hat{H}_{ZFS} = D_x \hat{S}_x^2 + D_y \hat{S}_y^2 + D_z \hat{S}_z^2. \quad (\text{A.1})$$

If we describe the x- and y-components of the D – tensor with its mean value and some deviation Δ

$$\begin{aligned} D_x \hat{S}_x^2 + D_y \hat{S}_y^2 &= \frac{D_x + D_y}{2} (\hat{S}_x^2 + \hat{S}_y^2) + \Delta \\ \Leftrightarrow \Delta &= \frac{D_x - D_y}{2} \hat{S}_x^2 + \frac{D_y - D_x}{2} \hat{S}_y^2 = \frac{D_x - D_y}{2} (\hat{S}_x^2 - \hat{S}_y^2). \end{aligned} \quad (\text{A.2})$$

Substitute this into Eq. (A.1)

$$\begin{aligned} \hat{H}_{ZFS} &= \frac{D_x + D_y}{2} (\hat{S}_x^2 + \hat{S}_y^2) + \frac{D_x - D_y}{2} (\hat{S}_x^2 - \hat{S}_y^2) + D_z \hat{S}_z^2 \\ &= \text{/\textbf{D} is traceless/} \\ &= -\frac{D_z}{2} (S(S+1) - \hat{S}_z^2) + \frac{D_x - D_y}{2} (\hat{S}_x^2 - \hat{S}_y^2) + D_z \hat{S}_z^2 \\ &= \frac{3}{2} D_z \left(\hat{S}_z^2 - \frac{S(S+1)}{3} \right) + \frac{D_x - D_y}{2} (\hat{S}_x^2 - \hat{S}_y^2). \end{aligned} \quad (\text{A.3})$$

From this expression we define the parameters

$$D = \frac{3}{2} D_z \text{ and } E = \frac{D_x - D_y}{2}. \quad (\text{A.4})$$

Symmetry expansion and coupling coefficients

From Eq. (2.26) we can read the coupling coefficients from the first- and second-order derivatives of the $\mathbf{D}$ tensor. The first-order coupling coefficients are

$$\frac{d\tilde{\mathbf{D}}}{dQ_i} = \begin{pmatrix} -1 & & \\ & -1 & \\ & & 2 \end{pmatrix} V_{00}^i \quad (\text{A.5})$$

$$\frac{d\tilde{\mathbf{D}}}{dQ_{l_x}} = \begin{pmatrix} -V_{+-}^l & & \sqrt{2}V_{0\pm}^l \\ & V_{+-}^l & \\ \sqrt{2}V_{0\pm}^l & & 0 \end{pmatrix} \quad (\text{A.6})$$

$$\frac{d\tilde{\mathbf{D}}}{dQ_{l_y}} = \begin{pmatrix} & V_{+-}^l & \\ V_{+-}^l & & \sqrt{2}V_{0\pm}^l \\ & \sqrt{2}V_{0\pm}^l & \end{pmatrix}. \quad (\text{A.7})$$

Second-order phonon perturbations will come in three combinations of $\{E_x, E_y\}$ pairs

$$A_1 \otimes A_1 : \frac{1}{2} \frac{d^2 \tilde{\mathbf{D}}}{dQ_i dQ_i} = \begin{pmatrix} -1 & & \\ & -1 & \\ & & 2 \end{pmatrix} V_{00}^{ii'} \quad (\text{A.8})$$

$$E_x \otimes E_x : \frac{1}{2} \frac{d^2 \tilde{\mathbf{D}}}{dQ_l dQ_l} = \begin{pmatrix} -V_{00}^{ll'} + V_{+-}^{ll'} & & \sqrt{2} V_{0\pm}^{ll'} \\ & -V_{00}^{ll'} - V_{+-}^{ll'} & \\ \sqrt{2} V_{0\pm}^{ll'} & & 2V_{00}^{ll'} \end{pmatrix} \quad (\text{A.9})$$

$$E_y \otimes E_y : \frac{1}{2} \frac{d^2 \tilde{\mathbf{D}}}{dQ_l dQ_l} = \begin{pmatrix} -V_{00}^{ll'} - V_{+-}^{ll'} & & \sqrt{2} V_{0\pm}^{ll'} \\ & -V_{00}^{ll'} + V_{+-}^{ll'} & \\ \sqrt{2} V_{0\pm}^{ll'} & & 2V_{00}^{ll'} \end{pmatrix} \quad (\text{A.10})$$

$$E_x \otimes E_y : \frac{1}{2} \frac{d^2 \tilde{\mathbf{D}}}{dQ_i dQ_i} = \begin{pmatrix} -V_{00}^{ll'} & 2V_{+-}^{ll'} & \\ 2V_{+-}^{ll'} & -V_{00}^{ll'} & \sqrt{2} V_{0\pm}^{ll'} \\ & \sqrt{2} V_{0\pm}^{ll'} & 2V_{00}^{ll'} \end{pmatrix} \quad (\text{A.11})$$

The index i will run over all A_1, A_2, E_x and E_y modes while the index l only runs over the E_x and E_y modes.

A.2 Time-dependent perturbation theory

In the presence of a time-dependent perturbation, the system Hamiltonian can be expressed as

$$\hat{H} = \hat{H}_0 + \hat{H}'(t), \quad (\text{A.12})$$

where $\hat{H}_0$ is the Hamiltonian for the unperturbed system with eigenvalues E_n and eigenstates $|n\rangle$, s.t. $\hat{H}_0 |n\rangle = E_n |n\rangle$ and $\hat{H}'(t)$ is the time-dependent perturbation. The time-dependent solution to the perturbed state can be expressed as

$$|\psi(t)\rangle = \sum_n a_n(t) e^{-iE_n t/\hbar} |n\rangle, \quad (\text{A.13})$$

where $a_n(t)$ is a probability amplitude. The probability of being in the state $|k\rangle$ at time t is $|a_k(t)|^2 = |\langle k | \psi(t) \rangle|^2$. Plugging Eq. (A.13) into the Schrödinger equation one obtains

$$\left(H_0 + H' - i\hbar \frac{\partial}{\partial t} \right) \sum_n a_n(t) e^{-iE_n t/\hbar} |n\rangle = 0. \quad (\text{A.14})$$

This can be expressed as a set of coupled partial differential equations (PDEs) by projecting onto the state $|k\rangle$

$$\frac{da_k(t)}{dt} = -\frac{i}{\hbar} \sum_n \langle k | H'(t) | n \rangle a_n(t) e^{-i(E_n - E_k)t/\hbar}. \quad (\text{A.15})$$

Finding exact solutions for these coupled PDEs can be difficult if there are many states $n = 1, 2, 3, \dots$, therefore we rewrite the PDEs to integral form

$$a_k(t) = a_k(0) - \frac{i}{\hbar} \sum_n \int_0^t \langle k | H'(t') | n \rangle a_n(t') e^{-i(E_n - E_k)t'/\hbar} dt' \quad (\text{A.16})$$

and find an iterative solution by iteratively substituting this expression for $a_k(t)$ into the right hand side and get

$$a_k(t) = a_k^{(0)} + a_k^{(1)} + a_k^{(2)} + \dots \quad (\text{A.17})$$

The amplitude of first order perturbation is then

$$a_k^{(1)}(t) = -\frac{i}{\hbar} \sum_n \int_0^t \langle k | H'(t') | n \rangle a_n^{(0)} e^{-i(E_n - E_k)t'/\hbar} dt'. \quad (\text{A.18})$$

Assuming the system is in initial state $|i\rangle$, the 0th order term is $a_n^{(0)} = a_n(t=0) = \delta_{ni}$ and the summation over n can be removed. Letting the final state be denoted as $|f\rangle$ we obtain

$$a_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t \langle f | H'(t') | i \rangle e^{i(E_k - E_n)t'/\hbar} dt', \quad (\text{A.19})$$

which is used to define Fermi's golden rule.

Fermi's golden rule

The transition rate $\Gamma_{f \leftarrow i}$ describes the rate at which transitions between a final state $|f\rangle$ and initial state $|i\rangle$ occurs. If we define $\omega_{fi} = (E_f - E_i) / \hbar$ we can write Eq. (A.19) as

$$a_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t \langle f | H'(t') | i \rangle e^{i\omega_{fi}t'} dt' \quad (\text{A.20})$$

and the probability of finding the system in state $|f\rangle$ at time t becomes

$$P_{fi}(t) = |a_f^{(1)}(t)|^2 = \frac{1}{\hbar^2} \left| \int_0^t \langle f | H'(t') | i \rangle e^{i\omega_{fi}t'} dt' \right|^2. \quad (\text{A.21})$$

For a harmonic perturbation of frequency ω we assume a perturbation Hamiltonian

$$\hat{H}'(t) = V (e^{-i\omega t} + e^{i\omega t}) \quad (\text{A.22})$$

where V is time-independent. Evaluated in Eq. (A.20), this yields

$$\begin{aligned} a_f^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t \langle f | V (e^{-i\omega t'} + e^{i\omega t'}) | i \rangle e^{i\omega_{fi}t'} dt' \\ &= -\frac{i}{\hbar} \int_0^t \langle f | V | i \rangle (e^{i(\omega_{fi}-\omega)t'} + e^{i(\omega_{fi}+\omega)t'}) dt' \\ &= -\langle f | V | i \rangle \frac{e^{i(\omega_{fi}-\omega)t} - 1}{\hbar(\omega_{fi} - \omega)} - \langle f | V | i \rangle \frac{e^{i(\omega_{fi}+\omega)t} - 1}{\hbar(\omega_{fi} + \omega)}. \end{aligned} \quad (\text{A.23})$$

If we look at the case of absorption i.e. $E_f \approx E_i + \hbar\omega$ we can neglect the second term since the major contribution comes from the first. This gives us the expression for the transition amplitude

$$P_{fi} = |a_f^{(1)}(t)|^2 = |\langle f | V | i \rangle|^2 \left| \frac{e^{i(\omega_{fi}-\omega)t} - 1}{\hbar(\omega_{fi} - \omega)} \right|^2 = |\langle f | V | i \rangle|^2 \frac{4 \sin^2 \frac{\omega_{fi}-\omega}{2} t}{\hbar^2(\omega_{fi} - \omega)^2} \quad (\text{A.24})$$

and

$$\begin{aligned} \Gamma_{f \leftarrow i}^{(1)} &= \lim_{t \rightarrow \infty} \frac{dP_{fi}}{dt} = \lim_{t \rightarrow \infty} |\langle f | V | i \rangle|^2 \frac{2 \sin \left((\omega_{fi} - \omega) t \right)}{\hbar^2(\omega_{fi} - \omega)} \\ &= \frac{2\pi}{\hbar} |\langle f | V | i \rangle|^2 \delta(E_f - E_i - \hbar\omega). \end{aligned} \quad (\text{A.25})$$

This is referred to as Fermi's golden rule, taken to first order in perturbation theory. In the case of emission, just replace $E_f - E_i - \hbar\omega$ with $E_f - E_i + \hbar\omega$.

Fermi's golden rule taken to second order in perturbation theory is derived in a similar manner and is given by the expression

$$\Gamma_{f \leftarrow i}^{(2),ll'} = \frac{2\pi}{\hbar} \left| \sum_n \frac{\langle f | V^{l'} | n \rangle \langle n | V^l | i \rangle}{E_n - E_i \pm \hbar\omega_l} \right|^2 \delta(E_f - E_i \pm \hbar(\omega_l \pm \omega_{l'})) \quad (\text{A.26})$$

where $|n\rangle$ is an intermediate state in the transition and l and l' represents two different perturbations. Since the final and initial state is independent of the intermediate states we can introduce the operator

$$G_{\pm} = \sum_n \frac{|n\rangle \langle n|}{E_n - E_i \pm \hbar\omega_l} \quad (\text{A.27})$$

which lets us express the second order transition rate as

$$\Gamma_{2,f \leftarrow i}^{(1)} = \frac{2\pi}{\hbar} \sum_{ll'} \left| \langle f | V^{l'} G_{\pm} V^l | i \rangle \right|^2 \delta(E_f - E_i \pm (\hbar\omega_{l'} \pm \hbar\omega_l)). \quad (\text{A.28})$$

Second-order phonon interactions

From Eq. (2.36) we can evaluate the second-order phonon eigenvalues as

$$\begin{aligned} \langle \mathbf{n}' | \hat{Q}_l \hat{Q}_{l'} | \mathbf{n} \rangle &= \frac{1}{2} \langle \mathbf{n}' | (\hat{b}_l + \hat{b}_l^\dagger)(\hat{b}_{l'} + \hat{b}_{l'}^\dagger) | \mathbf{n} \rangle \\ &= \langle \mathbf{n}' | \hat{b}_l \hat{b}_{l'} + \hat{b}_l \hat{b}_{l'}^\dagger + \hat{b}_l^\dagger \hat{b}_{l'} + \hat{b}_l^\dagger \hat{b}_{l'}^\dagger | \mathbf{n} \rangle \end{aligned} \quad (\text{A.29})$$

where the bold denotes the phonon state $|\mathbf{n}\rangle = |n_0, n_1, \dots, n_N\rangle$ and N is the number of phonon modes. This expression can be evaluated in four separate cases as the different phonon processes can not occur at the same time. The cases are also split up further depending on if it is the same phonon mode or a mix of two separate modes in the interaction. In the case of separate modes we have

$$\langle \mathbf{n}' | \hat{Q}_l \hat{Q}_{l'} | \mathbf{n} \rangle = \frac{1}{2} \begin{cases} \langle n_l + 1, n_{l'} + 1 | \hat{b}_l^\dagger \hat{b}_{l'}^\dagger | n_l, n_{l'} \rangle = \sqrt{n_l + 1} \sqrt{n_{l'} + 1} \\ \langle n_l - 1, n_{l'} - 1 | \hat{b}_l \hat{b}_{l'} | n_l, n_{l'} \rangle = \sqrt{n_l} \sqrt{n_{l'}} \\ \langle n_l + 1, n_{l'} - 1 | \hat{b}_l^\dagger \hat{b}_{l'} | n_l, n_{l'} \rangle = \sqrt{n_l + 1} \sqrt{n_{l'}} \\ \langle n_l - 1, n_{l'} + 1 | \hat{b}_l \hat{b}_{l'}^\dagger | n_l, n_{l'} \rangle = \sqrt{n_l} \sqrt{n_{l'} + 1} \end{cases}. \quad (\text{A.30})$$

For processes when $l = l'$ we get

$$\langle \mathbf{n}' | \hat{Q}_l \hat{Q}_l | \mathbf{n} \rangle = \frac{1}{2} \begin{cases} \langle n_l + 2 | \hat{b}_l^\dagger \hat{b}_l^\dagger | n_l \rangle = \sqrt{n_l + 1} \sqrt{n_l + 2} \\ \langle n_l - 2 | \hat{b}_l \hat{b}_l | n_l \rangle = \sqrt{n_l} \sqrt{n_l - 1} \\ \langle n_l | \hat{b}_l^\dagger \hat{b}_l | n_l \rangle = n_l \\ \langle n_l | \hat{b}_l \hat{b}_l^\dagger | n_l \rangle = n_l + 1 \end{cases}. \quad (\text{A.31})$$

In first-order perturbation theory and second-order phonon interactions, the phonon dimensionless coordinates are combined to fulfil the C_{3v} point group symmetry

$$\begin{cases} A_1 : \hat{R}_i^2, \\ E_x : \hat{X}_l^2 - \hat{Y}_l^2, \\ E_y : \hat{X}_l \hat{Y}_l + \hat{Y}_l \hat{X}_l. \end{cases} \quad (\text{A.32})$$

where $\hat{R}_i = (\hat{b}_i + \hat{b}_i^\dagger)/\sqrt{2}$ and $\{\hat{X}_l, \hat{Y}_l\} = (\hat{b}_{l_{\{x,y\}}} + \hat{b}_{l_{\{x,y\}}}^\dagger)/\sqrt{2}$.